

University of Minnesota
School of Physics and Astronomy

2026 Spring Physics 8502
General Relativity II
Assignment Solution

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Problem Set 1 due on Due Feb 9 at 11:59pm

Question 1

- (a) Consider the outer surface of the ergosphere for a Kerr black hole. Construct the normal vector to this surface and show that it is a 2-way surface.
- (b) Using the condition $n_\alpha n^\alpha = 0$ for a null surface, derive the equation for the critical 1-way surface for a time-independent axial symmetric surface. That is start with some $u(r, \theta)$ to derive the equation defining this surface. What is its physical interpretation.

Answer

(a)

Starting with the metric for a Kerr black hole in Boyer-Lindquist coordinates,

$$ds^2 = -\left(1 - \frac{2Mr}{\Sigma}\right) dt^2 - \frac{4Mar \sin^2 \theta}{\Sigma} dt d\phi + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \left(r^2 + a^2 + \frac{2Ma^2 r \sin^2 \theta}{\Sigma}\right) \sin^2 \theta d\phi^2, \quad (1)$$

where $\Sigma = r^2 + a^2 \cos^2 \theta$ and $\Delta = r^2 - 2Mr + a^2$. The outer surface of the ergosphere is defined by the condition $g_{tt} = 0$, which gives

$$1 - \frac{2Mr}{\Sigma} = 0 \implies r^2 - 2Mr + a^2 \cos^2 \theta = 0. \quad (2)$$

We can define a surface function $f(r, \theta) = r^2 - 2Mr + a^2 \cos^2 \theta = 0$. The normal vector to this surface is given by the gradient of f :

$$n_\mu = \partial_\mu f = \left(\frac{\partial f}{\partial t}, \frac{\partial f}{\partial r}, \frac{\partial f}{\partial \theta}, \frac{\partial f}{\partial \phi} \right) = (0, 2r - 2M, -2a^2 \cos \theta \sin \theta, 0). \quad (3)$$

To show that this is a 2-way surface, we need to compute the norm of the normal vector:

$$n_\mu n^\mu = g^{\mu\nu} n_\mu n_\nu. \quad (4)$$

We try to write out the matrix form of $g_{\mu\nu}$:

$$g_{\mu\nu} = \begin{pmatrix} -\left(1 - \frac{2Mr}{\Sigma}\right) & 0 & 0 & -\frac{2Mar \sin^2 \theta}{\Sigma} \\ 0 & \frac{\Sigma}{\Delta} & 0 & 0 \\ 0 & 0 & \Sigma & 0 \\ -\frac{2Mar \sin^2 \theta}{\Sigma} & 0 & 0 & \left(r^2 + a^2 + \frac{2Ma^2 r \sin^2 \theta}{\Sigma}\right) \sin^2 \theta \end{pmatrix}. \quad (5)$$

Since we are only interested in the rr and $\theta\theta$ components for the norm calculation, we can focus on those:

$$n_\mu n^\mu = g^{rr} n_r n_r + g^{\theta\theta} n_\theta n_\theta = g^{rr} (2r - 2M)^2 + g^{\theta\theta} (-2a^2 \cos \theta \sin \theta)^2. \quad (6)$$

Calculating g^{rr} and $g^{\theta\theta}$ from the inverse metric, we find:

$$g^{rr} = \frac{\Delta}{\Sigma}, \quad g^{\theta\theta} = \frac{1}{\Sigma}. \quad (7)$$

Substituting these back into the norm expression:

$$n_\mu n^\mu = \frac{\Delta}{\Sigma} (2r - 2M)^2 + \frac{1}{\Sigma} (-2a^2 \cos \theta \sin \theta)^2 \quad (8)$$

$$= \frac{4}{\Sigma} [\Delta (r - M)^2 + a^4 \cos^2 \theta \sin^2 \theta] \quad (9)$$

$$= \frac{4}{\Sigma} [(r^2 - 2Mr + a^2)(r - M)^2 + a^4 \cos^2 \theta \sin^2 \theta]. \quad (10)$$

We have $r^2 - 2Mr = -a^2 \cos^2 \theta$, and $(r - M)^2 = r^2 - 2Mr + M^2 = -a^2 \cos^2 \theta + M^2$ on the ergosphere, so substituting this in:

$$n_\mu n^\mu = \frac{4}{\Sigma} [a^2 \sin^2 \theta (r - M)^2 + a^4 \cos^2 \theta \sin^2 \theta] = \frac{4a^2 \sin^2 \theta}{\Sigma} [(r - M)^2 + a^2 \cos^2 \theta]. \quad (11)$$

Since for a Kerr black hole $a \neq 0$ and $\sin \theta \neq 0$ except at the poles, we have $n_\mu n^\mu > 0$ almost everywhere on the ergosphere, indicating that the normal vector is spacelike. Therefore, the ergosphere is a 2-way surface.

(b)

For a time-independent axially symmetric surface, we can define the surface function as $u(r, \theta) = 0$. The normal vector to this surface is given by:

$$n_\mu = \partial_\mu u = \left(0, \frac{\partial u}{\partial r}, \frac{\partial u}{\partial \theta}, 0 \right). \quad (12)$$

Using the condition for a null surface, $n_\mu n^\mu = 0$, we have:

$$n_\mu n^\mu = g^{rr} \left(\frac{\partial u}{\partial r} \right)^2 + g^{\theta\theta} \left(\frac{\partial u}{\partial \theta} \right)^2 = 0. \quad (13)$$

If we plug the $g^{rr} = \frac{\Delta}{\Sigma}$ and $g^{\theta\theta} = \frac{1}{\Sigma}$ from the Kerr metric, we get:

$$\frac{\Delta}{\Sigma} \left(\frac{\partial u}{\partial r} \right)^2 + \frac{1}{\Sigma} \left(\frac{\partial u}{\partial \theta} \right)^2 = 0 \quad (14)$$

$$\implies \Delta \left(\frac{\partial u}{\partial r} \right)^2 + \left(\frac{\partial u}{\partial \theta} \right)^2 = 0. \quad (15)$$

If we want this equation to hold, we can have following condition:

1. $\Delta = 0$ and $\frac{\partial u}{\partial \theta} = 0$. This corresponds to the event horizon of the Kerr black hole, which is a 1-way surface.
2. $\frac{\partial u}{\partial r} = 0$ and $\frac{\partial u}{\partial \theta} = 0$. This would imply that u is constant, which is not a valid surface definition.

Thus, the critical 1-way surface is defined by $\Delta = 0$ and $\frac{\partial u}{\partial \theta} = 0$. That is, $u(r, \theta) = u(r)$ only, and the surface is located at the event horizon of the Kerr black hole. The condition $\Delta = 0$ corresponds to the event horizon, which is a null surface that allows one-way passage of information. $\square$

Question 2

Show that

$$(L_X T)^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} = (\nabla_X T)^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} - \sum_i T^{a_1 \dots j \dots a_r}_{b_1 b_2 \dots b_s} X^a_i + \sum_i T^{a_1 a_2 \dots a_r}_{b_1 \dots j \dots b_s} X^j_{;b_i} \quad (16)$$

Answer

In the course, we have already know the definition of the Lie derivative of a tensor field T along a vector field X :

$$(L_X T)^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} = X^c \partial_c T^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} - \sum_i T^{a_1 \dots j \dots a_r}_{b_1 b_2 \dots b_s} X^a_i + \sum_i T^{a_1 a_2 \dots a_r}_{b_1 \dots j \dots b_s} X^j_{;b_i}. \quad (17)$$

On the other hand, the covariant derivative of T along X is given by:

$$(\nabla_X T)^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} = X^c \partial_c T^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} + \sum_i \Gamma^{a_i}_{cj} T^{a_1 \dots j \dots a_r}_{b_1 b_2 \dots b_s} X^c - \sum_i \Gamma^j_{cb_i} T^{a_1 a_2 \dots a_r}_{b_1 \dots j \dots b_s} X^c. \quad (18)$$

Subtracting the two expressions, we get:

$$(L_X T)^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} - (\nabla_X T)^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} = - \sum_i T^{a_1 \dots j \dots a_r}_{b_1 b_2 \dots b_s} X^a_i + \sum_i T^{a_1 a_2 \dots a_r}_{b_1 \dots j \dots b_s} X^j_{;b_i} \quad (19)$$

$$- \sum_i \Gamma^{a_i}_{cj} T^{a_1 \dots j \dots a_r}_{b_1 b_2 \dots b_s} X^c + \sum_i \Gamma^j_{cb_i} T^{a_1 a_2 \dots a_r}_{b_1 \dots j \dots b_s} X^c \quad (20)$$

$$= - \sum_i T^{a_1 \dots j \dots a_r}_{b_1 b_2 \dots b_s} X^a_i + \sum_i T^{a_1 a_2 \dots a_r}_{b_1 \dots j \dots b_s} X^j_{;b_i}. \quad (21)$$

We have used the definition of the covariant derivative of a vector field, $X^a_{;b} = X^a_b + \Gamma^a_{cb} X^c$, to rewrite the terms involving the Christoffel symbols. Thus, we have shown that:

$$(L_X T)^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} = (\nabla_X T)^{a_1 a_2 \dots a_r}_{b_1 b_2 \dots b_s} - \sum_i T^{a_1 \dots j \dots a_r}_{b_1 b_2 \dots b_s} X^a_i + \sum_i T^{a_1 a_2 \dots a_r}_{b_1 \dots j \dots b_s} X^j_{;b_i}. \quad (22)$$

□

Question 3

Consider an arbitrary unit vector, X transported along a latitude line on the surface of the sphere. Use $ds^2 = d\theta^2 + \sin^2 \theta d\phi^2$, and recall that there are only 2 distinct Γ 's.

- (a) Show the behavior of the angle between that vector and the tangent vector, T , which generates the isometry (latitude line) as that vector is parallel transported along the latitude line.
- (b) Do the same assuming the vector is Lie transported. That is instead of $\nabla_T X = 0$, assume that $L_T X = 0$.

Answer

Before we start, let's write down the non-zero Christoffel symbols for the metric on the surface of the sphere:

$$\Gamma_{\phi\phi}^{\theta} = -\sin \theta \cos \theta, \quad \Gamma_{\theta\phi}^{\phi} = \Gamma_{\phi\theta}^{\phi} = \cot \theta. \quad (23)$$

(a)

The metric on the surface of the sphere is given by $ds^2 = d\theta^2 + \sin^2 \theta d\phi^2$. The tangent vector to the latitude line is $T = \frac{\partial}{\partial \phi}$, which generates the isometry. The unit vector X can be expressed in terms of the coordinate basis as $X = X^{\theta} \frac{\partial}{\partial \theta} + X^{\phi} \frac{\partial}{\partial \phi}$. The angle α between X and T is given by:

$$\cos \alpha = \frac{g(X, T)}{\sqrt{g(X, X)g(T, T)}}. \quad (24)$$

Since $T = \frac{\partial}{\partial \phi}$, we have $g(T, T) = g_{\phi\phi} = \sin^2 \theta$. The inner product $g(X, T)$ is given by:

$$g(X, T) = g_{\phi\phi} X^{\phi} = \sin^2 \theta X^{\phi}. \quad (25)$$

The norm of X is given by:

$$g(X, X) = g_{\theta\theta} (X^{\theta})^2 + g_{\phi\phi} (X^{\phi})^2 = (X^{\theta})^2 + \sin^2 \theta (X^{\phi})^2. \quad (26)$$

For convenience, we set the length of X to be 1, Thus, the cosine of the angle is:

$$\cos \alpha = \frac{\sin^2 \theta X^{\phi}}{\sqrt{((X^{\theta})^2 + \sin^2 \theta (X^{\phi})^2) \sin^2 \theta}} = \frac{\sin \theta X^{\phi}}{\sqrt{(X^{\theta})^2 + \sin^2 \theta (X^{\phi})^2}} \quad (27)$$

$$= \frac{\sin \theta X^{\phi}}{\sqrt{1}} = \sin \theta X^{\phi}. \quad (28)$$

To find how the angle changes as X is parallel transported along the latitude line, we need to solve the parallel transport equation $\nabla_T X = 0$. This gives us:

$$\nabla_T X^\theta = T^\phi \partial_\phi X^\theta + \Gamma_{\phi\phi}^\theta T^\phi X^\phi = 0, \quad (29)$$

$$\nabla_T X^\phi = T^\phi \partial_\phi X^\phi + \Gamma_{\phi\theta}^\phi T^\phi X^\theta + \Gamma_{\phi\phi}^\phi T^\phi X^\phi = 0. \quad (30)$$

Substituting $T^\phi = 1$ and the Christoffel symbols, we get:

$$\partial_\phi X^\theta - \sin \theta \cos \theta X^\phi = 0, \quad (31)$$

$$\partial_\phi X^\phi + \cot \theta X^\theta = 0. \quad (32)$$

Now we can try to solve these coupled differential equations. From the first equation, we can express X^ϕ in terms of X^θ :

$$X^\phi = \frac{1}{\sin \theta \cos \theta} \partial_\phi X^\theta. \quad (33)$$

Substituting this into the second equation gives us a second-order differential equation for X^θ :

$$\partial_\phi \left(\frac{1}{\sin \theta \cos \theta} \partial_\phi X^\theta \right) + \cot \theta X^\theta = 0 \quad (34)$$

$$\implies \frac{1}{\sin \theta \cos \theta} \partial_\phi^2 X^\theta + \cot \theta X^\theta = 0 \quad (35)$$

$$\implies \partial_\phi^2 X^\theta + \cot \theta \sin \theta \cos \theta X^\theta = 0 \quad (36)$$

$$\implies \partial_\phi^2 X^\theta + \cos^2 \theta X^\theta = 0. \quad (37)$$

The general solution to this equation is:

$$X^\theta = A \cos((\cos \theta)\phi) + B \sin((\cos \theta)\phi) = C \sin((\cos \theta)\phi + \delta), \quad (38)$$

where A , B , C , and δ are constants determined by the initial conditions. Substituting this back into the expression for X^ϕ , we get:

$$X^\phi = \frac{1}{\sin \theta \cos \theta} \partial_\phi X^\theta = \frac{C}{\sin \theta} \cos((\cos \theta)\phi + \delta). \quad (39)$$

Now we can compute the angle α as a function of ϕ :

$$\cos \alpha(\phi) = \sin \theta X^\phi = C \cos((\cos \theta)\phi + \delta). \quad (40)$$

Thus, the angle between the vector X and the tangent vector T oscillates as X is parallel transported along the latitude line, with a frequency determined by $\cos \theta$.

(b)

Now we assume that the vector X is Lie transported along the latitude line, which means that $L_T X = 0$. The Lie derivative of X along T is given by:

$$(L_T X)^\theta = T^\phi \partial_\phi X^\theta - X^\phi \partial_\phi T^\theta = \partial_\phi X^\theta, \quad (41)$$

$$(L_T X)^\phi = T^\phi \partial_\phi X^\phi - X^\theta \partial_\phi T^\phi = \partial_\phi X^\phi. \quad (42)$$

Setting $L_T X = 0$ gives us:

$$\partial_\phi X^\theta = 0, \quad (43)$$

$$\partial_\phi X^\phi = 0. \quad (44)$$

This means that both X^θ and X^ϕ are constants along the latitude line. Therefore, the angle α between X and T is also constant:

$$\cos \alpha = \sin \theta X^\phi = \text{constant}. \quad (45)$$

Thus, when the vector X is Lie transported along the latitude line, the angle between X and the tangent vector T remains constant, in contrast to the oscillatory behavior observed in the case of parallel transport. $\square$

Problem Set 2 due on Due Feb 18 at 11:59pm

Question 1

Consider 2 objects of mass m_1 and m_2 with separation a , orbiting about a common center of mass. Find the change in the period ($\dot{\tau}/\tau$) due to gravitational radiation. Assume the above result is valid as $a \rightarrow 0$, find the time to go from $a = a_0$ to $a = 0$.

Answer

We first write down the radiation power of the system:

$$P = \frac{dE}{dt} = -\frac{G}{5c^5} \left\langle \ddot{I}_{ij} \ddot{I}_{ij} - \frac{1}{3} (\ddot{I}_{kk})^2 \right\rangle \quad (46)$$

$$= -\frac{G}{5} \left\langle \ddot{I}_{ij} \ddot{I}_{ij} - \frac{1}{3} (\ddot{I}_{kk})^2 \right\rangle \quad (47)$$

where I_{ij} is the quadrupole moment of the system. For a binary system, we can express I_{ij} in terms of the reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$ and the separation vector $\mathbf{r}$ between the two masses:

$$I_{ij} = \mu(r_i r_j - \frac{1}{3} r^2 \delta_{ij}) \quad (48)$$

For a circular orbit, the separation vector can be expressed as $\mathbf{r}(t) = a(\cos(\omega t), \sin(\omega t), 0)$, where ω is the angular frequency of the orbit. The third time derivative of I_{ij} can be calculated as follows:

$$\ddot{I}_{ij} = \mu \left(\frac{d^3}{dt^3} (r_i r_j) - \frac{1}{3} \frac{d^3}{dt^3} (r^2 \delta_{ij}) \right) \quad (49)$$

Calculating the third time derivative of $r_i r_j$ and $r^2 \delta_{ij}$, we find:

$$\frac{d^3}{dt^3} (r_i r_j) = \frac{d^3}{dt^3} \begin{pmatrix} a^2 \cos^2(\omega t) & a^2 \cos(\omega t) \sin(\omega t) & 0 \\ a^2 \cos(\omega t) \sin(\omega t) & a^2 \sin^2(\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (50)$$

$$= \frac{d^3}{dt^3} \begin{pmatrix} \frac{a^2}{2} (1 + \cos(2\omega t)) & \frac{a^2}{2} \sin(2\omega t) & 0 \\ \frac{a^2}{2} \sin(2\omega t) & \frac{a^2}{2} (1 - \cos(2\omega t)) & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (51)$$

$$= 4a^2 \omega^3 \begin{pmatrix} \sin(2\omega t) & -\cos(2\omega t) & 0 \\ -\cos(2\omega t) & -\sin(2\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (52)$$

$$\frac{d^3}{dt^3} (r^2 \delta_{ij}) = \frac{d^3}{dt^3} (a^2 \delta_{ij}) = 0 \quad (53)$$

Substituting these results back into the expression for $\ddot{I}_{ij}$, we get:

$$\ddot{I}_{ij} = 4\mu a^2 \omega^3 \begin{pmatrix} \sin(2\omega t) & -\cos(2\omega t) & 0 \\ -\cos(2\omega t) & -\sin(2\omega t) & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad (54)$$

Now we can calculate the power radiated by the system:

$$P = \frac{dE}{dt} = -\frac{G}{5} \left\langle \ddot{I}_{ij} \ddot{I}_{ij} - \frac{1}{3} (\ddot{I}_{kk})^2 \right\rangle \quad (55)$$

$$= -\frac{G}{5} \langle 16\mu^2 a^4 \omega^6 (2\sin^2(2\omega t) + 2\cos^2(2\omega t)) \rangle \quad (56)$$

$$= -\frac{32G}{5} \mu^2 a^4 \omega^6. \quad (57)$$

By Kepler's third law, we have $\omega^2 = \frac{G(m_1+m_2)}{a^3}$, $\mu = \frac{m_1 m_2}{m_1+m_2}$, which allows us to express the power in terms of the separation a :

$$P = -\frac{32G^4}{5} \frac{\mu^2 (m_1 + m_2)^3}{a^5} \quad (58)$$

$$= -\frac{32G^4}{5} \frac{m_1^2 m_2^2 (m_1 + m_2)}{a^5} \quad (59)$$

The energy of the system is given by the sum of the kinetic and potential energy:

$$E = -\frac{Gm_1 m_2}{2a} \quad (60)$$

The rate of change of the energy is equal to the power radiated:

$$\frac{dE}{dt} = \frac{Gm_1 m_2}{2a^2} \frac{da}{dt} = -\frac{32G^4}{5} \frac{m_1^2 m_2^2 (m_1 + m_2)}{a^5} \quad (61)$$

Solving for $\frac{da}{dt}$, we find:

$$\frac{da}{dt} = -\frac{64G^3}{5} \frac{m_1 m_2 (m_1 + m_2)}{a^3} \quad (62)$$

Now, we can write down the period of the orbit $\tau = \frac{2\pi}{\omega}$:

$$\tau = 2\pi \sqrt{\frac{a^3}{G(m_1 + m_2)}} = \frac{2\pi}{\sqrt{G(m_1 + m_2)}} a^{3/2} \quad (63)$$

Taking the time derivative of the period, we get:

$$\frac{d\tau}{dt} = \frac{3\pi}{\sqrt{G(m_1 + m_2)}} a^{1/2} \frac{da}{dt} = -\frac{192\pi G^{5/2}}{5} \frac{m_1 m_2 (m_1 + m_2)^{1/2}}{a^{5/2}} \quad (64)$$

Finally, we can express the change in the period as:

$$\frac{\dot{\tau}}{\tau} = \frac{d\tau/dt}{\tau} = -\frac{96G^3}{5} \frac{m_1 m_2}{a^4} \quad (65)$$

To find the time it takes for the separation to go from $a = a_0$ to $a = 0$, we can integrate the expression for $\frac{da}{dt}$:

$$t = \int dt = \int_{a_0}^0 \frac{da}{\frac{da}{dt}} = \int_{a_0}^0 -\frac{5}{64G^3} \frac{a^3}{m_1 m_2 (m_1 + m_2)} da = \frac{5a_0^4}{256G^3 m_1 m_2 (m_1 + m_2)}. \quad (66)$$

□

Question 2

Problem 7.6 in Carroll. Two object of mass M have a head on collision at $(0,0,0,0)$. In the distant past, $t \rightarrow -\infty$, the mass mass started at $x \rightarrow \pm\infty$ with zero velocity.

- (a) Using Newtonian theory show that $x(t) = \pm(\frac{9}{8}GMt^2)^{1/3}$.
- (b) For what separations is the Newtonian approximation reasonable?
- (c) Calculate h_{xx}^{TT} at $(0, R, 0)$.
- (d) For the same problem, calculate the total energy radiated in the collision.

Answer

(a)

Start with energy conservation:

$$2\frac{1}{2}M\dot{x}^2 - \frac{GM^2}{2x} = 0 \quad (67)$$

$$\implies \dot{x} = \sqrt{\frac{GM}{2x}} \quad (68)$$

Separating variables and integrating, we get:

$$\int x^{1/2} dx = \sqrt{\frac{GM}{2}} \int dt \quad (69)$$

$$\implies \frac{2}{3}x^{3/2} = \sqrt{\frac{GM}{2}}t + C, \quad \text{where } C = 0 \text{ since } x = 0 \text{ at } t = 0 \quad (70)$$

$$\implies x^{3/2} = \frac{3}{2}\sqrt{\frac{GM}{2}}t = \sqrt{\frac{9}{8}GMt} \quad (71)$$

$$\implies x(t) = \pm \left(\frac{9}{8}GMt^2\right)^{1/3} \quad (72)$$

(b)

The Newtonian approximation is reasonable when the gravitational field is weak and the velocities are much less than the speed of light. For the speed, we can calculate $\dot{x}$ from the expression we derived in

part (a):

$$\dot{x} = \frac{d}{dt} \left(\frac{9}{8} GM t^2 \right)^{1/3} = \frac{2}{3} \left(\frac{9}{8} GM \right)^{1/3} t^{-1/3} \ll c = 1 \quad (73)$$

$$\Rightarrow t \gg \left(\frac{2}{3} \left(\frac{9}{8} GM \right)^{1/3} \right)^3 = \frac{8}{27} GM \quad (74)$$

$$\Rightarrow x(t) \gg \left(\frac{9}{8} GM \left(\frac{8}{27} GM \right)^2 \right)^{1/3} = \frac{2}{3} GM \quad (75)$$

For the gravitational field, we can calculate the gravitational potential Φ at the position of one of the masses:

$$\Phi = -\frac{GM}{2x} \quad (76)$$

The Newtonian approximation is reasonable when $|\Phi| \ll 1$, which implies:

$$x \gg \frac{GM}{2} \quad (77)$$

(c)

By equation (7.140) in Carroll, the transverse-traceless part of the metric perturbation is given by:

$$h_{ij}^{TT} = \frac{2G}{R} \frac{d^2}{dt^2} I_{ij}(t_r), \quad (78)$$

where I_{ij} is the quadrupole moment of the system and $t_r = t - R$ is the retarded time. The quadrupole moment can be calculated as:

$$I_{ij} = \sum_a m_a (x_a^i x_a^j - \frac{1}{3} r_a^2 \delta_{ij}). \quad (79)$$

Since we want to calculate h_{xx}^{TT} , we need to find I_{xx} . For the two masses, we have:

$$I_{xx} = M(x^2 - \frac{1}{3}r^2) + M((-x)^2 - \frac{1}{3}r^2) = \frac{4}{3}Mx^2 = \frac{4}{3}M \left(\frac{9}{8} GM t^2 \right)^{2/3} \quad (80)$$

Taking the second time derivative, we get:

$$\frac{d^2}{dt^2} I_{xx} = \frac{4}{3} M \frac{d^2}{dt^2} \left(\frac{9}{8} GM t^2 \right)^{2/3} = \frac{4}{3} M \left(\frac{9}{8} GM \right)^{2/3} \frac{d^2}{dt^2} t^{4/3} \quad (81)$$

$$= \frac{4}{3} M \left(\frac{9}{8} GM \right)^{2/3} \frac{4}{9} t^{-2/3} = \frac{16}{27} M \left(\frac{9}{8} GM \right)^{2/3} t^{-2/3} \quad (82)$$

Substituting this back into the expression for h_{xx}^{TT} , we get:

$$h_{xx}^{TT} = \frac{2G}{R} \frac{16}{27} M \left(\frac{9}{8} GM \right)^{2/3} t^{-2/3} = \frac{32GM}{27R} \left(\frac{9}{8} GM \right)^{2/3} (t - R)^{-2/3}, \quad (83)$$

where we have replaced t with the retarded time $t_r = t - R$.

(d)

The total energy radiated in the collision can be calculated using the quadrupole formula for gravitational radiation:

$$E = \frac{G}{5} \int_{-\infty}^{\infty} \left\langle \ddot{I}_{ij} \ddot{I}_{ij} - \frac{1}{3} (\ddot{I}_{kk})^2 \right\rangle dt \quad (84)$$

Since we have already calculated I_{xx} , we can find $\ddot{I}_{xx}$ by taking the third time derivative:

$$\ddot{I}_{xx} = \frac{d}{dt} \frac{d^2}{dt^2} I_{xx} = \frac{d}{dt} \left(\frac{16}{27} M \left(\frac{9}{8} GM \right)^{2/3} t^{-2/3} \right) = -\frac{32}{81} M \left(\frac{9}{8} GM \right)^{2/3} t^{-5/3} \quad (85)$$

Since the quadrupole moment is traceless, we have $\ddot{I}_{kk} = 0$. Therefore, the energy radiated can be expressed as:

$$I_{yy} = I_{zz} = -\frac{1}{2} I_{xx}, \quad \ddot{I}_{yy} = \ddot{I}_{zz} = -\frac{1}{2} \ddot{I}_{xx} \quad (86)$$

Back to the expression for the energy radiated, we have:

$$E = \frac{G}{5} \int_{-\infty}^{t_0} \left\langle \ddot{I}_{xx}^2 + 2 \ddot{I}_{yy}^2 - \frac{1}{3} (\ddot{I}_{xx} + 2 \ddot{I}_{yy})^2 \right\rangle dt \quad (87)$$

$$= \frac{G}{5} \int_{-\infty}^{t_0} \left\langle \ddot{I}_{xx}^2 + 2 \left(-\frac{1}{2} \ddot{I}_{xx} \right)^2 - \frac{1}{3} (\ddot{I}_{xx} - \ddot{I}_{xx})^2 \right\rangle dt \quad (88)$$

$$= \frac{G}{5} \int_{-\infty}^{t_0} \left\langle \ddot{I}_{xx}^2 + 2 \cdot \frac{1}{4} \ddot{I}_{xx}^2 - 0 \right\rangle dt \quad (89)$$

$$= \frac{G}{5} \int_{-\infty}^{t_0} \left\langle \frac{3}{2} \ddot{I}_{xx}^2 \right\rangle dt = \frac{3G}{10} \int_{-\infty}^{t_0} \left(-\frac{32}{81} M \left(\frac{9}{8} GM \right)^{2/3} t^{-5/3} \right)^2 dt \quad (90)$$

$$= \frac{3G}{10} \frac{1024}{6561} M^2 \left(\frac{9}{8} GM \right)^{4/3} \int_{-\infty}^{t_0} t^{-10/3} dt = \frac{3G}{10} \frac{1024}{6561} M^2 \left(\frac{9}{8} GM \right)^{4/3} \cdot \frac{3}{7} t_0^{-7/3} \quad (91)$$

$$= \frac{512}{10935} GM^2 \left(\frac{9}{8} GM \right)^{4/3} t_0^{-7/3}, \quad (92)$$

where t_0 is the time at which the collision occurs. Since the collision occurs at $t = 0$, we can take the limit as $t_0 \rightarrow 0$ to find the total energy radiated:

$$E = \lim_{t_0 \rightarrow 0} \frac{512}{10935} GM^2 \left(\frac{9}{8} GM \right)^{4/3} t_0^{-7/3} = \infty \quad (93)$$

This result indicates that an infinite amount of energy is radiated in the collision, which is a consequence

of the idealized nature of the problem. In reality, the energy radiated would be finite due to various factors such as the **finite size** of the masses and the presence of other forces that would come into play during the collision. $\square$

Question 3

A ball of mass $m = 100$ g is thrown into the vacuum above the earth (ie. neglect all effects of air resistance), which produces a UNIFORM gravitational field ($g = 10^3$ cm s²) with a velocity $v_0 = 10^3$ cm/s. Normally, this ball would rise to a height of $h = v_0^2/2g = 500$ cm. However, the ball will be a source of gravitational radiation and won't go quite so high. Find Δh . How does $\Delta h/h$ depend on v_0 ?

Now suppose the ball will fall back and elastically bounce. Left alone, the ball will eventually come to rest. How long will it take.

Answer

The power radiated by the ball due to gravitational radiation can be calculated using the quadrupole formula:

$$P = \frac{G}{5} \left\langle \ddot{I}_{ij} \ddot{I}_{ij} - \frac{1}{3} (\ddot{I}_{kk})^2 \right\rangle \quad (94)$$

For a ball moving in a uniform gravitational field, the quadrupole moment can be expressed as:

$$I_{ij} = m(x_i x_j - \frac{1}{3} r^2 \delta_{ij}) \quad (95)$$

Since the ball is moving vertically, we can express its position as $x(t) = (0, 0, z(t))$, where $z(t)$ is the height of the ball at time t . The quadrupole moment can then be simplified to:

$$I_{zz} = m(z^2 - \frac{1}{3} z^2) = \frac{2}{3} m z^2 \quad (96)$$

Since the quadrupole moment is traceless, we have $I_{xx} = I_{yy} = -\frac{1}{2} I_{zz}$. Therefore, the second time derivative of the quadrupole moment can be calculated as:

$$\ddot{I}_{zz} = \frac{d^3}{dt^3} I_{zz} = \frac{d^3}{dt^3} \left(\frac{2}{3} m z^2 \right) \quad (97)$$

$$= \frac{2}{3} m \frac{d^2}{dt^2} (2z\dot{z}) = \frac{4}{3} m \frac{d}{dt} (z\ddot{z} + \dot{z}^2) \quad (98)$$

$$= \frac{4m}{3} \frac{d}{dt} (z(-g) + \dot{z}^2) = \frac{4m}{3} (-g\dot{z} + 2\dot{z}\ddot{z}) = \frac{4m}{3} (-g\dot{z} + 2\dot{z}(-g)) = -4mg\dot{z} = -4mgv \quad (99)$$

Since $I_{xx} = I_{yy} = -\frac{1}{2} I_{zz}$, we have $\ddot{I}_{xx} = \ddot{I}_{yy} = -\frac{1}{2} \ddot{I}_{zz}$. Substituting these results back into the expression for the power, we get:

$$P = \frac{G}{5} \left\langle \ddot{I}_{zz}^2 + 2\ddot{I}_{xx}^2 - \frac{1}{3} (\ddot{I}_{zz} + 2\ddot{I}_{xx})^2 \right\rangle = \frac{G}{5} \left\langle \ddot{I}_{zz}^2 + 2 \cdot \frac{1}{4} \ddot{I}_{zz}^2 - 0 \right\rangle = \frac{3G}{10} \left\langle \ddot{I}_{zz}^2 \right\rangle \quad (100)$$

$$= \frac{3G}{10} \left\langle (-4mgv)^2 \right\rangle = \frac{24Gm^2g^2}{5} v^2 \quad (101)$$

The energy radiated by the ball can be calculated by integrating the power over time, and we also restore the c^5 in the denominator:

$$\Delta E = \int P dt = \int \frac{24Gm^2g^2}{5c^5} v^2 dt = \frac{24Gm^2g^2}{5c^5} \int v^2 dt \quad (102)$$

$$= \frac{24Gm^2g^2}{5c^5} \int v^2 \frac{dv}{-g} = -\frac{24Gm^2g}{5c^5} \int v^2 dv = -\frac{24Gm^2g}{5c^5} \cdot \frac{v_0^3}{3} = -\frac{8Gm^2gv_0^3}{5c^5} \quad (103)$$

The change in height Δh can be calculated by equating the energy radiated to the change in potential energy:

$$\Delta E = mg\Delta h \implies \Delta h = \frac{\Delta E}{mg} = -\frac{8Gm^2gv_0^3}{5c^5} \quad (104)$$

The ratio $\Delta h/h$ can be expressed as:

$$\frac{\Delta h}{h} = \frac{-\frac{8Gm^2gv_0^3}{5c^5}}{\frac{v_0^2}{2g}} = -\frac{16Gm^2g^2v_0}{5c^5} \quad (105)$$

This shows that the ratio $\Delta h/h$ is proportional to the initial velocity v_0 of the ball.

Now, we know the due to the radiation, we have $\Delta E = -\frac{8Gm^2gv_0^3}{5c^5}$, which means the ball loses energy at a rate of $\frac{dE}{dt}$:

$$\frac{dE}{dt} = -\frac{8Gm^2gv_0^3}{5c^5} \cdot \frac{1}{t_{\text{total}}} = -\frac{8Gm^2gv_0^3}{5c^5} \cdot \frac{1}{2t_{\text{up}}}, \quad (106)$$

where t_{total} is the total time for the ball to go up and come back down, and t_{up} is the time it takes for the ball to reach its maximum height. Since the ball is thrown upwards with an initial velocity of v_0 , we can calculate t_{up} using the equation of motion:

$$v = v_0 - gt_{\text{up}} = 0 \implies t_{\text{up}} = \frac{v_0}{g} \quad (107)$$

Substituting this back into the expression for $\frac{dE}{dt}$,

$$\frac{dE}{dt} = -\frac{8Gm^2gv_0^3}{5c^5} \cdot \frac{1}{2 \cdot \frac{v_0}{g}} = -\frac{4Gm^2g^2v_0^2}{5c^5} \quad (108)$$

Note that the energy of the ball at its maximum height is given by:

$$E = mgh = mg \frac{v_0^2}{2g} = \frac{1}{2}mv_0^2 \implies v_0^2 = \frac{2E}{m}, \quad (109)$$

Hence, we can express $\frac{dE}{dt}$ in terms of the energy E :

$$\frac{dE}{dt} = -\frac{4Gm^2g^2}{5c^5} \cdot \frac{2E}{m} = -\frac{8Gmg^2}{5c^5}E \quad (110)$$

This is a first-order linear differential equation, the solution to which is given by:

$$E(t) = E_0 e^{-\frac{8Gmg^2}{5c^5}t} \quad (111)$$

where E_0 is the initial energy of the ball at its maximum height. The ball will come to rest when its energy approaches zero, which occurs as $t \rightarrow \infty$.

□

Problem Set 3 due on Due March 2 at 11:59pm

Question 1

Show that the ratio of the mean density to the central density in a polytrope is

$$\frac{\bar{\rho}}{\rho(0)} = 3 \frac{|\theta'(\xi_1)|}{\xi_1}. \quad (112)$$

Answer

We start with the definition of the mean density $\bar{\rho}$, which is given by the total mass M divided by the volume V of the star:

$$\bar{\rho} = \frac{M}{V}. \quad (113)$$

The total mass M can be expressed as an integral of the density $\rho(r)$ over the volume of the star:

$$M = \int_0^R 4\pi r^2 \rho(r) dr, \quad (114)$$

where R is the radius of the star. The volume V of the star is given by:

$$V = \frac{4}{3}\pi R^3. \quad (115)$$

Substituting these expressions into the definition of $\bar{\rho}$, we get:

$$\bar{\rho} = \frac{\int_0^R 4\pi r^2 \rho(r) dr}{\frac{4}{3}\pi R^3} = \frac{3}{R^3} \int_0^R r^2 \rho(r) dr. \quad (116)$$

Now, we can express the density $\rho(r)$ in terms of the dimensionless variable ξ and the function $\theta(\xi)$, which is related to the density by:

$$\rho(r) = \rho(0)\theta^n(\xi), \quad (117)$$

where n is the polytropic index. The variable ξ is defined as:

$$\xi = \frac{r}{a}, \quad (118)$$

where a is a scaling factor. The radius of the star corresponds to ξ_1 , where $\theta(\xi_1) = 0$. Substituting these expressions into the integral for $\bar{\rho}$, we get:

$$\bar{\rho} = \frac{3}{R^3} \int_0^R r^2 \rho(0) \theta^n\left(\frac{r}{a}\right) dr. \quad (119)$$

Changing the variable of integration from r to ξ , we have:

$$\bar{\rho} = \frac{3}{R^3} \int_0^{\xi_1} (a\xi)^2 \rho(0) \theta^n(\xi) a d\xi = \frac{3a^3 \rho(0)}{R^3} \int_0^{\xi_1} \xi^2 \theta^n(\xi) d\xi. \quad (120)$$

Since $R = a\xi_1$, we can substitute this into the expression for $\bar{\rho}$:

$$\bar{\rho} = \frac{3a^3 \rho(0)}{(a\xi_1)^3} \int_0^{\xi_1} \xi^2 \theta^n(\xi) d\xi = \frac{3\rho(0)}{\xi_1^3} \int_0^{\xi_1} \xi^2 \theta^n(\xi) d\xi. \quad (121)$$

Now, we can use the Lane-Emden equation, which relates $\theta(\xi)$ to its derivative $\theta'(\xi)$:

$$\frac{1}{\xi^2} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) = -\theta^n. \quad (122)$$

Integrating this equation from 0 to ξ_1 , we get:

$$\int_0^{\xi_1} \xi^2 \theta^n(\xi) d\xi = - \int_0^{\xi_1} \frac{d}{d\xi} \left(\xi^2 \frac{d\theta}{d\xi} \right) d\xi = - \left[\xi^2 \frac{d\theta}{d\xi} \right]_0^{\xi_1} = -\xi_1^2 \theta'(\xi_1). \quad (123)$$

Substituting this back into the expression for $\bar{\rho}$, we get:

$$\bar{\rho} = \frac{3\rho(0)}{\xi_1^3} (-\xi_1^2 \theta'(\xi_1)) = 3 \frac{|\theta'(\xi_1)|}{\xi_1} \rho(0). \quad (124)$$

Finally, we can express the ratio of the mean density to the central density as:

$$\frac{\bar{\rho}}{\rho(0)} = 3 \frac{|\theta'(\xi_1)|}{\xi_1}. \quad (125)$$

□

Question 2

Show that the mean kinetic energy of an electron in a degenerate gas is $\frac{3}{5}E'_F$ in the non-relativistic case and $\frac{3}{4}E_F$ in the relativistic case. Here $E'_F = E_F - m_e \approx \frac{p_F^2}{2m_e}$.

Answer

The definition of the mean kinetic energy $\langle K \rangle$ is given by the total kinetic energy of the electrons divided by the total number of electrons. In a degenerate electron gas, the electrons fill up all available energy states up to the Fermi energy E_F . In the non-relativistic case, the kinetic energy of an electron with momentum p is given by:

$$K = \frac{p^2}{2m_e}. \quad (126)$$

The Fermi energy E_F is related to the Fermi momentum p_F by:

$$E_F = \frac{p_F^2}{2m_e}. \quad (127)$$

The mean kinetic energy can be calculated by integrating the kinetic energy over all occupied states up to the Fermi momentum:

$$\langle K \rangle = \frac{1}{N} \int_0^{p_F} \frac{p^2}{2m_e} g(p) dp, \quad (128)$$

where $g(p)$ is the density of states in momentum space and N is the total number of electrons. The density of states in three dimensions is given by:

$$g(p) = \frac{V}{(2\pi)^3} 4\pi p^2, \quad (129)$$

where V is the volume of the system. The total number of electrons N can be calculated by integrating the density of states up to the Fermi momentum:

$$N = \int_0^{p_F} g(p) dp = \frac{V}{(2\pi)^3} \int_0^{p_F} 4\pi p^2 dp = \frac{V}{(2\pi)^3} \frac{4\pi p_F^3}{3}. \quad (130)$$

Substituting the expressions for $g(p)$ and N into the expression for $\langle K \rangle$, we get:

$$\langle K \rangle = \frac{1}{N} \int_0^{p_F} \frac{p^2}{2m_e} \frac{V}{(2\pi)^3} 4\pi p^2 dp = \frac{1}{N} \frac{V}{(2\pi)^3} \int_0^{p_F} \frac{p^4}{2m_e} 4\pi dp. \quad (131)$$

Evaluating the integral, we have:

$$\langle K \rangle = \frac{1}{N} \frac{V}{(2\pi)^3} \frac{4\pi}{2m_e} \int_0^{p_F} p^4 dp = \frac{1}{N} \frac{V}{(2\pi)^3} \frac{4\pi}{2m_e} \frac{p_F^5}{5}. \quad (132)$$

Substituting the expression for N , we get:

$$\langle K \rangle = \frac{1}{\frac{V}{(2\pi)^3} \frac{4\pi p_F^3}{3}} \frac{V}{(2\pi)^3} \frac{4\pi}{2m_e} \frac{p_F^5}{5} = \frac{3}{5} \frac{p_F^2}{2m_e} = \frac{3}{5} E'_F. \quad (133)$$

In the relativistic case, the kinetic energy of an electron with momentum p is given by:

$$K = \sqrt{p^2 c^2 + m_e^2 c^4} - m_e c^2 \approx pc, \quad (134)$$

where we have used the approximation for high momentum. The Fermi energy E_F is related to the Fermi momentum p_F by:

$$E_F = \sqrt{p_F^2 c^2 + m_e^2 c^4} - m_e c^2 \approx p_F c. \quad (135)$$

The mean kinetic energy can be calculated by integrating the kinetic energy over all occupied states up to the Fermi momentum:

$$\langle K \rangle = \frac{1}{N} \int_0^{p_F} pcg(p)dp, \quad (136)$$

where $g(p)$ and N are the same as in the non-relativistic case. Substituting the expressions for $g(p)$ and N , we get:

$$\langle K \rangle = \frac{1}{N} \int_0^{p_F} pc \frac{V}{(2\pi)^3} 4\pi p^2 dp = \frac{1}{N} \frac{V}{(2\pi)^3} 4\pi c \int_0^{p_F} p^3 dp. \quad (137)$$

Evaluating the integral, we have:

$$\langle K \rangle = \frac{1}{N} \frac{V}{(2\pi)^3} 4\pi c \frac{p_F^4}{4} = \frac{1}{N} \frac{V}{(2\pi)^3} \pi c p_F^4. \quad (138)$$

Substituting the expression for N , we get:

$$\langle K \rangle = \frac{1}{\frac{V}{(2\pi)^3} \frac{4\pi p_F^3}{3}} \frac{V}{(2\pi)^3} \pi c p_F^4 = \frac{3}{4} p_F c = \frac{3}{4} E_F. \quad (139)$$

□

Question 3

For a pure ideal neutron gas, solve the coupled differential equations for mass and p_F to find the central density for which the total mass is maximum. Then provide the mass (max) mass and radius at that value. Plot the mass as a function of central density showing both regions of stability and instability.

Hint: Use the last 2 pages of notes in StStr2.pdf, this converts the O.V. equation to variables x and y . You will need expressions for $\bar{p}$ and $\bar{\rho}$ for a ideal fermi gas of neutrons. Then you will need to integrate (numerically) the OV equation and the equation for $d\bar{M}/dx$. Each initial value of y (when $x = 0$) corresponds to a central density. The radius of the star will correspond to the value of x when the density goes to 0. You can make a table (say from y_{init} between 0.35 and 1 and determine the mass of the star for each y_{init} to find the value of y_{init} when M is maximized).

Answer

The Oppenheimer-Volkoff (OV) equations is given by:

$$\frac{dP}{dr} = -\frac{G(M(r) + 4\pi r^3 P/c^2)(\rho + P/c^2)}{r(r - 2GM(r)/c^2)}, \quad (140)$$

$$\frac{dM}{dr} = 4\pi r^2 \rho, \quad (141)$$

where P is the pressure, ρ is the density, $M(r)$ is the mass enclosed within radius r , and G is the gravitational constant. For a pure ideal neutron gas, we can express the pressure and density in terms of the Fermi momentum p_F as follows:

$$P = \frac{1}{3\pi^2} \int_0^{p_F} \frac{p^4}{\sqrt{p^2 c^2 + m_n^2 c^4}} dp, \quad (142)$$

$$\rho = \frac{1}{\pi^2} \int_0^{p_F} p^2 \sqrt{p^2 c^2 + m_n^2 c^4} dp, \quad (143)$$

where m_n is the mass of a neutron. We can introduce dimensionless variables x and y defined as:

$$x = \frac{r}{R_0}, \quad (144)$$

$$y = \frac{p_F}{m_n c}, \quad (145)$$

where R_0 is a characteristic radius. The OV equations can be rewritten in terms of these dimensionless variables as:

$$\frac{dP}{dx} = -\frac{G(M(x) + 4\pi R_0^3 x^3 P/c^2)(\rho + P/c^2)}{x(R_0 x - 2GM(x)/c^2)}, \quad (146)$$

$$\frac{dM}{dx} = 4\pi R_0^3 x^2 \rho. \quad (147)$$

Also, we can express the pressure and density in terms of y as:

$$P(y) = \frac{1}{3\pi^2} \int_0^y \frac{(m_n c y')^4}{\sqrt{(m_n c y')^2 c^2 + m_n^2 c^4}} m_n c dy' \quad (148)$$

$$= \frac{m_n^4 c^5}{3\pi^2} \int_0^y \frac{y'^4}{\sqrt{y'^2 + 1}} dy', \quad (149)$$

$$\rho(y) = \frac{1}{\pi^2} \int_0^y (m_n c y')^2 \sqrt{(m_n c y')^2 c^2 + m_n^2 c^4} m_n c dy' \quad (150)$$

$$= \frac{m_n^4 c^5}{\pi^2} \int_0^y y'^2 \sqrt{y'^2 + 1} dy'. \quad (151)$$

By chain rule, we can express the derivatives in terms of y :

$$\frac{dP}{dx} = \frac{dP}{dy} \frac{dy}{dx} \quad (152)$$

$$= \frac{m_n^4 c^5}{3\pi^2} \frac{y^4}{\sqrt{y^2 + 1}} \frac{dy}{dx}. \quad (153)$$

Hence, the OV equations can be rewritten as:

$$\frac{m_n^4 c^5}{3\pi^2} \frac{y^4}{\sqrt{y^2 + 1}} \frac{dy}{dx} = - \frac{G(M(x) + 4\pi R_0^3 x^3 P(y)/c^2)(\rho(y) + P(y)/c^2)}{x(R_0 x - 2GM(x)/c^2)}. \quad (154)$$

We can simplify the first equation to find an expression for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = - \frac{3\pi^2 G(M(x) + 4\pi R_0^3 x^3 P(y)/c^2)(\rho(y) + P(y)/c^2)}{m_n^4 c^5 x(R_0 x - 2GM(x)/c^2) \frac{y^4}{\sqrt{y^2 + 1}}}. \quad (155)$$

Now we have following coupled differential equations:

$$\frac{dy}{dx} = - \frac{3\pi^2 G(M(x) + 4\pi R_0^3 x^3 P(y)/c^2)(\rho(y) + P(y)/c^2)}{m_n^4 c^5 x(R_0 x - 2GM(x)/c^2) \frac{y^4}{\sqrt{y^2 + 1}}}, \quad (156)$$

$$\frac{dM}{dx} = 4\pi R_0^3 x^2 \rho(y), \quad (157)$$

$$P(y) = \frac{m_n^4 c^5}{3\pi^2 \hbar^3} \int_0^y \frac{y'^4}{\sqrt{y'^2 + 1}} dy', \quad (158)$$

$$\rho(y) = \frac{m_n^4 c^5}{\pi^2 \hbar^3} \int_0^y y'^2 \sqrt{y'^2 + 1} dy'. \quad (159)$$

We restore the physical constants $\hbar$ in the expressions for pressure and density to ensure the correct units. We define the following constants to simplify the equations:

$$\epsilon = \frac{m_n^4 c^5}{3\pi^2 \hbar^3}, \quad (160)$$

$$R_0 = \frac{c^2}{\sqrt{G\epsilon}}, \quad R_0^2 = \frac{c^4}{G\epsilon} \quad (161)$$

$$\mu = \frac{R_0 c^2}{G}, \quad (162)$$

$$\tilde{m} = M/\mu, \quad M = \tilde{m}\mu = \tilde{m} \frac{R_0 c^2}{G}, \quad (163)$$

$$\tilde{R} = R_0 x, \quad (164)$$

Then we can rewrite the equations as:

$$P(y) = \epsilon \int_0^y \frac{y'^4}{\sqrt{y'^2 + 1}} dy' = \epsilon f(y), \quad (165)$$

$$\rho(y) = 3\epsilon \int_0^y y'^2 \sqrt{y'^2 + 1} dy' = 3\epsilon g(y)/c^2, \quad (166)$$

$$\frac{dM}{dx} = \frac{d\tilde{m}}{dx} \mu = \frac{d\tilde{m}}{dx} \frac{R_0 c^2}{G} = 4\pi R_0^3 x^2 \rho(y) \implies \frac{d\tilde{m}}{dx} = 4\pi \frac{G R_0^2}{c^2} x^2 \rho(y) \quad (167)$$

$$\implies \frac{d\tilde{m}}{dx} = 4\pi \frac{G R_0^2}{c^2} x^2 3\epsilon g(y)/c^2 = 12\pi \frac{G R_0^2 \epsilon}{c^4} x^2 g(y) = 12\pi x^2 g(y) \quad (168)$$

$$\frac{dy}{dx} = - \frac{3\pi^2 G (M(x) + 4\pi R_0^3 x^3 P(y)/c^2) (\rho(y) + P(y)/c^2)}{m_n^4 c^5 x (R_0 x - 2GM(x)/c^2) \frac{y^4}{\sqrt{y^2+1}}} \quad (169)$$

$$= - \frac{3\pi^2 G (\tilde{m} \frac{R_0 c^2}{G} + 4\pi R_0^3 x^3 \epsilon f(y)/c^2) (3\epsilon g(y)/c^2 + \epsilon f(y)/c^2)}{m_n^4 c^5 x (R_0 x - 2G\tilde{m} \frac{R_0 c^2}{G}/c^2) \frac{y^4}{\sqrt{y^2+1}}} \quad (170)$$

The first term:

$$(\tilde{m} \frac{R_0 c^2}{G} + 4\pi R_0^3 x^3 \epsilon f(y)/c^2) = \frac{R_0 c^2}{G} (\tilde{m} + 4\pi \frac{G R_0^2 \epsilon}{c^4} x^3 f(y)) \quad (171)$$

$$= \frac{R_0 c^2}{G} (\tilde{m} + 4\pi x^3 f(y)) \quad (172)$$

The second term:

$$(3\epsilon g(y)/c^2 + \epsilon f(y)/c^2) = \epsilon(3g(y) + f(y))/c^2 \quad (173)$$

The denominator:

$$m_n^4 c^5 x(R_0 x - 2G\tilde{m} \frac{R_0 c^2}{G}/c^2) = (3\pi^2 \epsilon) R_0 x(x - 2\tilde{m}) \quad (174)$$

Putting everything together, we have:

$$\frac{dy}{dx} = - \frac{3\pi^2 G(\tilde{m} \frac{R_0 c^2}{G} + 4\pi R_0^3 x^3 \epsilon f(y)/c^2)(3\epsilon g(y) + \epsilon f(y)/c^2)}{m_n^4 c^5 x(R_0 x - 2G\tilde{m} \frac{R_0 c^2}{G}/c^2) \frac{y^4}{\sqrt{y^2+1}}} \quad (175)$$

$$= - \frac{3\pi^2 G \frac{R_0 c^2}{G} (\tilde{m} + 4\pi x^3 f(y)) \epsilon(3g(y) + f(y)/c^2)}{(3\pi^2 \epsilon) R_0 x(x - 2\tilde{m}) \frac{y^4}{\sqrt{y^2+1}}} \quad (176)$$

$$= - \frac{3\pi^2 G \frac{R_0 c^2}{G} (\tilde{m} + 4\pi x^3 f(y)) \epsilon(3g(y) + f(y))/c^2}{(3\pi^2 \epsilon) R_0 x(x - 2\tilde{m}) \frac{y^4}{\sqrt{y^2+1}}} \quad (177)$$

$$= - \frac{(\tilde{m} + 4\pi x^3 f(y))(3g(y) + f(y))}{x(x - 2\tilde{m}) \frac{y^4}{\sqrt{y^2+1}}} \quad (178)$$

In summary, we have the following coupled differential equations:

$$\frac{dy}{dx} = - \frac{(\tilde{m} + 4\pi x^3 f(y))(3g(y) + f(y))}{x(x - 2\tilde{m}) \frac{y^4}{\sqrt{y^2+1}}}, \quad (179)$$

$$\frac{d\tilde{m}}{dx} = 12\pi x^2 g(y), \quad (180)$$

$$f(y) = \int_0^y \frac{y'^4}{\sqrt{y'^2 + 1}} dy', \quad (181)$$

$$g(y) = \int_0^y y'^2 \sqrt{y'^2 + 1} dy'. \quad (182)$$

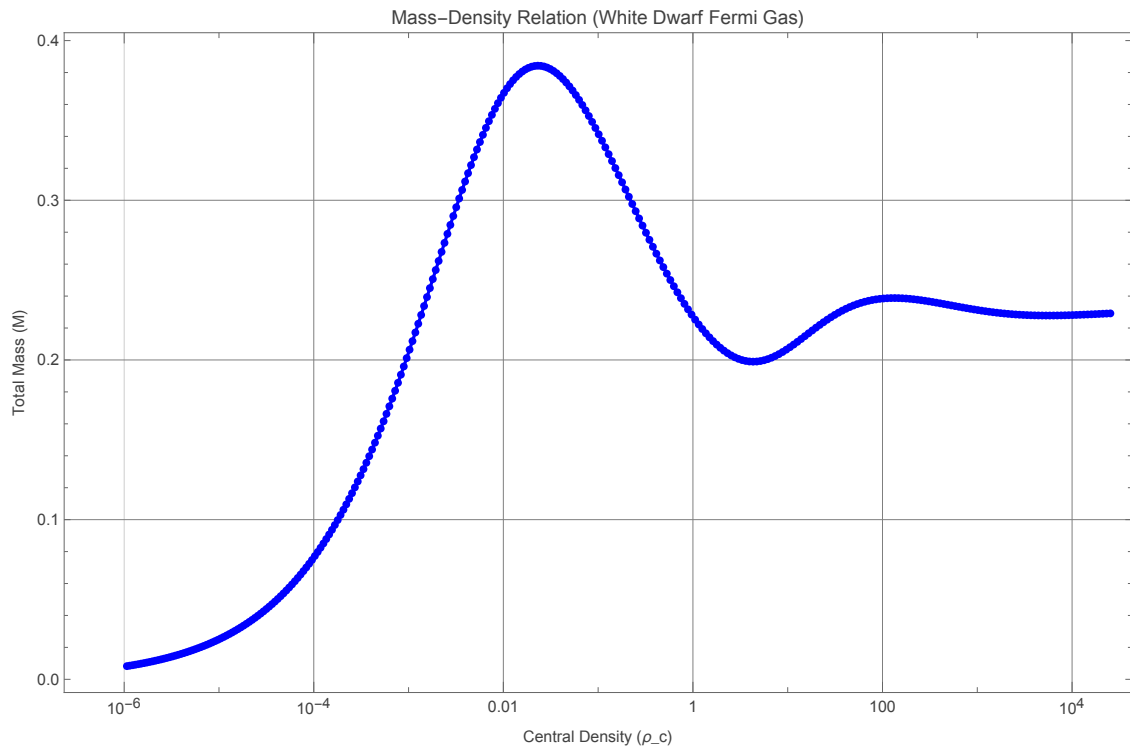


Figure 1: Mass as a function of central density. The maximum mass occurs at the peak of the curve, which corresponds to the critical central density. The region to the left of the peak is stable, while the region to the right is unstable.

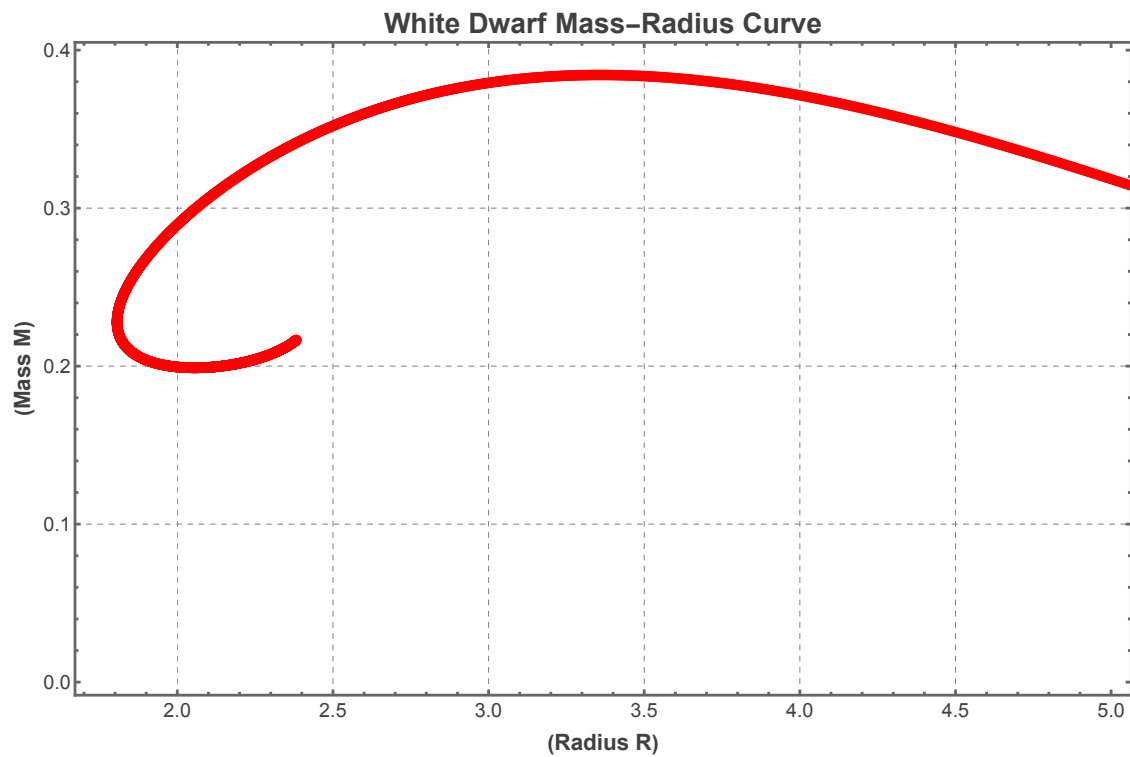


Figure 2: Mass as a function of radius.

Problem Set 4 due on Due March 18 at 11:59pm

Question 1

Consider a 2D space with negative curvature. Show that the embedding into Euclidean space does not yield a space with constant curvature for either hyperboloid:

(a) $x^2 + y^2 - z^2 = R^2$

(b) $x^2 + y^2 - z^2 = -R^2$

Find the Ricci scalar for both cases. Then show the case (b) embedded in a non-Euclidean space does give a space of constant curvature and find the Ricci scalar. Also, what happens in case (a) for a non-Euclidean embedding?

Answer

(a)

For the hyperboloid $x^2 + y^2 - z^2 = R^2$, we can parametrize the surface using cylindrical coordinates (z, ϕ) :

$$x = \sqrt{R^2 + z^2} \cos \phi, \quad (183)$$

$$y = \sqrt{R^2 + z^2} \sin \phi, \quad (184)$$

$$z = z. \quad (185)$$

The original metric in the embedding space is given by

$$ds^2 = dx^2 + dy^2 + dz^2. \quad (186)$$

Then we have

$$dx = \frac{z}{\sqrt{R^2 + z^2}} \cos \phi dz - \sqrt{R^2 + z^2} \sin \phi d\phi, \quad (187)$$

$$dy = \frac{z}{\sqrt{R^2 + z^2}} \sin \phi dz + \sqrt{R^2 + z^2} \cos \phi d\phi, \quad (188)$$

$$dz = dz. \quad (189)$$

Now we can compute the metric on the surface by substituting these expressions into the original metric:

$$ds^2 = dx^2 + dy^2 + dz^2 \quad (190)$$

$$= \left(\frac{z^2}{R^2 + z^2} + 1 \right) dz^2 + (R^2 + z^2) d\phi^2 \quad (191)$$

$$= \frac{R^2 + 2z^2}{R^2 + z^2} dz^2 + (R^2 + z^2) d\phi^2. \quad (192)$$

Then, we can say in the coordinates (z, ϕ) , the metric is given by

$$g_{zz} = \frac{R^2 + 2z^2}{R^2 + z^2}, \quad (193)$$

$$g_{\phi\phi} = R^2 + z^2, \quad (194)$$

$$g_{z\phi} = g_{\phi z} = 0. \quad (195)$$

After calculation in Mathematica (see the attached file), we can find the Ricci scalar is given by

$$\mathcal{R} = -\frac{2R^2}{(R^2 + 2z^2)^2}. \quad (196)$$

(b)

For the hyperboloid $x^2 + y^2 - z^2 = -R^2$, we can parametrize the surface using cylindrical coordinates (z, ϕ) :

$$x = \sqrt{z^2 - R^2} \cos \phi, \quad (197)$$

$$y = \sqrt{z^2 - R^2} \sin \phi, \quad (198)$$

$$z = z. \quad (199)$$

The original metric in the embedding space is given by

$$ds^2 = dx^2 + dy^2 + dz^2. \quad (200)$$

Then we have

$$dx = \frac{z}{\sqrt{z^2 - R^2}} \cos \phi dz - \sqrt{z^2 - R^2} \sin \phi d\phi, \quad (201)$$

$$dy = \frac{z}{\sqrt{z^2 - R^2}} \sin \phi dz + \sqrt{z^2 - R^2} \cos \phi d\phi, \quad (202)$$

$$dz = dz. \quad (203)$$

Now we can compute the metric on the surface by substituting these expressions into the original metric:

$$ds^2 = dx^2 + dy^2 + dz^2 \quad (204)$$

$$= \left(\frac{z^2}{z^2 - R^2} + 1 \right) dz^2 + (z^2 - R^2) d\phi^2 \quad (205)$$

$$= \frac{2z^2 - R^2}{z^2 - R^2} dz^2 + (z^2 - R^2) d\phi^2. \quad (206)$$

Then, we can say in the coordinates (z, ϕ) , the metric is given by

$$g_{zz} = \frac{2z^2 - R^2}{z^2 - R^2}, \quad (207)$$

$$g_{\phi\phi} = z^2 - R^2, \quad (208)$$

$$g_{z\phi} = g_{\phi z} = 0. \quad (209)$$

After calculation in Mathematica (see the attached file), we can find the Ricci scalar is given by

$$\mathcal{R} = \frac{2R^2}{(2z^2 - R^2)^2}. \quad (210)$$

Now, we can see that the Ricci scalar is not a constant, which means the curvature is not constant. However, if we embed the hyperboloid into a non-Euclidean space with metric

$$ds^2 = dx^2 + dy^2 - dz^2, \quad (211)$$

we can find the metric on the surface is given by

$$ds^2 = dx^2 + dy^2 - dz^2 \quad (212)$$

$$= \left(\frac{z^2}{z^2 - R^2} - 1 \right) dz^2 + (z^2 - R^2) d\phi^2 \quad (213)$$

$$= \frac{R^2}{z^2 - R^2} dz^2 + (z^2 - R^2) d\phi^2. \quad (214)$$

Then, we can say in the coordinates (z, ϕ) , the metric is given by

$$g_{zz} = \frac{R^2}{z^2 - R^2}, \quad (215)$$

$$g_{\phi\phi} = z^2 - R^2, \quad (216)$$

$$g_{z\phi} = g_{\phi z} = 0. \quad (217)$$

After calculation in Mathematica (see the attached file), we can find the Ricci scalar is given by

$$\mathcal{R} = -\frac{2}{R^2}. \quad (218)$$

We can see that the Ricci scalar is a constant, which means the curvature is constant.

(a)

For the case (a), if we embed the hyperboloid into a non-Euclidean space with metric

$$ds^2 = dx^2 + dy^2 - dz^2, \quad (219)$$

we can find the metric on the surface is given by

$$ds^2 = dx^2 + dy^2 - dz^2 \quad (220)$$

$$= \left(\frac{z^2}{R^2 + z^2} - 1 \right) dz^2 + (R^2 + z^2) d\phi^2 \quad (221)$$

$$= - \frac{R^2}{R^2 + z^2} dz^2 + (R^2 + z^2) d\phi^2. \quad (222)$$

Then, we can say in the coordinates (z, ϕ) , the metric is given by

$$g_{zz} = - \frac{R^2}{R^2 + z^2}, \quad (223)$$

$$g_{\phi\phi} = R^2 + z^2, \quad (224)$$

$$g_{z\phi} = g_{\phi z} = 0. \quad (225)$$

After calculation in Mathematica (see the attached file), we can find the Ricci scalar is given by

$$\mathcal{R} = \frac{2}{R^2}. \quad (226)$$

We can see that the Ricci scalar is a constant, which means the curvature is constant.

□

Question 2

Consider the metric

$$g_{\mu\nu} = C_{\mu\nu} + \frac{K}{1 - KC_{ab}x^ax^b}C_{\mu\lambda}C_{\nu\sigma}x^\lambda x^\sigma \quad (227)$$

- (a) Derive the inverse metric $g^{\mu\nu}$.
- (b) Derive the Christoffel symbols Γ_{bc}^a .
- (c) Derive solutions for the equations of motions when $K = +1, 0, -1$.

Answer

(a)

To derive the inverse metric $g^{\mu\nu}$, we start with the given metric:

$$g_{\mu\nu} = C_{\mu\nu} + \frac{K}{1 - KC_{ab}x^ax^b}C_{\mu\lambda}C_{\nu\sigma}x^\lambda x^\sigma. \quad (228)$$

We can rewrite this as:

$$g_{\mu\nu} = C_{\mu\nu} + \frac{K}{1 - KC_{ab}x^ax^b}(C_{\mu\lambda}x^\lambda)(C_{\nu\sigma}x^\sigma). \quad (229)$$

Let us define $x_\mu = C_{\mu\lambda}x^\lambda$. Also, we can define $C_{ab}x^ax^b = x^\rho x_\rho$ since $x^\rho = C^{\rho\lambda}x_\lambda$. Then we can express the metric as:

$$g_{\mu\nu} = C_{\mu\nu} + \frac{K}{1 - Kx^\rho x_\rho}x_\mu x_\nu. \quad (230)$$

To find the inverse metric $g^{\mu\nu}$, we can assume it has the form:

$$g^{\mu\nu} = C^{\mu\nu} + \alpha x^\mu x^\nu, \quad (231)$$

where α is a constant to be determined. We need to satisfy the condition:

$$g^{\mu\rho}g_{\rho\nu} = \delta^\mu{}_\nu. \quad (232)$$

Substituting the expressions for $g_{\mu\nu}$ and $g^{\nu\rho}$, we get:

$$(C^{\mu\rho} + \alpha x^\mu x^\rho)(C_{\rho\nu} + \frac{K}{1 - Kx^\rho x_\rho}x_\rho x_\nu) = \delta^\mu{}_\nu. \quad (233)$$

Expanding this expression, we have:

$$C^{\mu\rho}C_{\rho\nu} + \frac{K}{1 - Kx^\rho x_\rho} C^{\mu\rho} x_\rho x_\nu + \alpha x^\mu x^\rho C_{\rho\nu} + \alpha \frac{K}{1 - Kx^\rho x_\rho} x^\mu x^\rho x_\rho x_\nu = \delta^\mu{}_\nu. \quad (234)$$

Since $C^{\mu\rho}C_{\rho\nu} = \delta^\mu{}_\nu$, we can simplify this to:

$$0 = \frac{K}{1 - Kx^\rho x_\rho} C^{\mu\rho} x_\rho x_\nu + \alpha x^\mu x_\nu + \alpha \frac{K}{1 - Kx^\rho x_\rho} x^\mu x^\rho x_\rho x_\nu \quad (235)$$

$$= \frac{K}{1 - Kx^\rho x_\rho} x^\mu x_\nu + \alpha x^\mu x_\nu + \alpha \frac{K}{1 - Kx^\rho x_\rho} x^\mu (x^\rho x_\rho) x_\nu \quad (236)$$

$$= \left(\frac{K}{1 - Kx^\rho x_\rho} + \alpha + \alpha \frac{K}{1 - Kx^\rho x_\rho} (x^\rho x_\rho) \right) x^\mu x_\nu. \quad (237)$$

Hence, we can solve for α :

$$\alpha \left(1 + \frac{Kx^\rho x_\rho}{1 - Kx^\rho x_\rho} \right) = - \frac{K}{1 - Kx^\rho x_\rho} \quad (238)$$

$$\implies \alpha \frac{1}{1 - Kx^\rho x_\rho} = - \frac{K}{1 - Kx^\rho x_\rho} \quad (239)$$

$$\implies \alpha = -K. \quad (240)$$

Thus, the inverse metric is given by:

$$g^{\mu\nu} = C^{\mu\nu} - Kx^\mu x^\nu. \quad (241)$$

(b)

First, we need to compute the derivatives of the metric components. The metric is given by:

$$g_{\mu\nu} = C_{\mu\nu} + \frac{K}{1 - Kx^\rho x_\rho} x_\mu x_\nu = C_{\mu\nu} + \frac{K}{1 - KC_{ab}x^a x^b} C_{\mu\lambda} C_{\nu\sigma} x^\lambda x^\sigma = C_{\mu\nu} + f(x) x_\mu x_\nu, \quad (242)$$

where we have defined $f(x) = \frac{K}{1 - Kx^\rho x_\rho}$. Then, we can compute the derivatives of the metric components:

$$\partial_\lambda g_{\mu\nu} = \underbrace{\partial_\lambda C_{\mu\nu}}_{=0} + \partial_\lambda f(x) x_\mu x_\nu + f(x) \partial_\lambda (x_\mu x_\nu) \quad (243)$$

$$= \partial_\lambda f(x) x_\mu x_\nu + f(x) (\partial_\lambda x_\mu x_\nu + x_\mu \partial_\lambda x_\nu) \quad (244)$$

$$= \partial_\lambda f(x) x_\mu x_\nu + f(x) (\delta_{\lambda\mu} x_\nu + x_\mu \delta_{\lambda\nu}). \quad (245)$$

Also, the $\partial_\lambda f(x)$ can be computed as:

$$\partial_\lambda f(x) = \partial_\lambda \left(\frac{K}{1 - Kx^\rho x_\rho} \right) \quad (246)$$

$$= \frac{K^2 \partial_\lambda (x^\rho x_\rho)}{(1 - Kx^\rho x_\rho)^2} \quad (247)$$

$$= \frac{K^2 \partial_\lambda (C^{\rho\sigma} x_\sigma x_\rho)}{(1 - Kx^\rho x_\rho)^2} \quad (248)$$

$$= \frac{K^2 (C^{\rho\sigma} \delta_{\lambda\sigma} x_\rho + C^{\rho\sigma} x_\sigma \delta_{\lambda\rho})}{(1 - Kx^\rho x_\rho)^2} \quad (249)$$

$$= \frac{2K^2 x_\lambda}{(1 - Kx^\rho x_\rho)^2}. \quad (250)$$

Hence the derivatives of the metric components can be expressed as:

$$\partial_\lambda g_{\mu\nu} = \frac{2K^2 x_\lambda}{(1 - Kx^\rho x_\rho)^2} x_\mu x_\nu + f(x) (\delta_{\lambda\mu} x_\nu + \delta_{\lambda\nu} x_\mu). \quad (251)$$

Now we can compute the Christoffel symbols using the formula:

$$\Gamma_{bc}^a = \frac{1}{2} g^{ad} (\partial_b g_{cd} + \partial_c g_{bd} - \partial_d g_{bc}). \quad (252)$$

Substituting the expressions for g^{ad} and $\partial_\lambda g_{\mu\nu}$, we can compute the Christoffel symbols:

$$\Gamma_{bc}^a = \frac{1}{2} (C^{ad} - Kx^a x^d) \left(\frac{2K^2 x_b}{(1 - Kx^\rho x_\rho)^2} x_c x_d + f(x) (\delta_{bc} x_d + \delta_{bd} x_c) \right) \quad (253)$$

$$+ \frac{2K^2 x_c}{(1 - Kx^\rho x_\rho)^2} x_b x_d + f(x) (\delta_{cb} x_d + \delta_{cd} x_b) - \frac{2K^2 x_d}{(1 - Kx^\rho x_\rho)^2} x_b x_c - f(x) (\delta_{db} x_c + \delta_{dc} x_b) \quad (254)$$

$$= \frac{1}{2} (C^{ad} - Kx^a x^d) \left(\frac{2K^2}{(1 - Kx^\rho x_\rho)^2} x_b x_c x_d + 2 \frac{K}{1 - Kx^\rho x_\rho} \delta_{bc} x_d \right) \quad (255)$$

$$= \frac{K}{1 - Kx^\rho x_\rho} (C^{ad} - Kx^a x^d) \left(\frac{K}{1 - Kx^\rho x_\rho} x_b x_c x_d + \delta_{bc} x_d \right) \quad (256)$$

$$= \frac{K}{1 - Kx^\rho x_\rho} \left(\frac{K}{1 - Kx^\rho x_\rho} x_b x_c x^a + \delta_{bc} x^a - \frac{K^2}{1 - Kx^\rho x_\rho} x^a x^d x_d x_b x_c - K \delta_{bc} x^a x^d x_d \right) \quad (257)$$

$$= \frac{K}{1 - Kx^\rho x_\rho} x^a \left(\frac{K}{1 - Kx^\rho x_\rho} x_b x_c (1 - Kx^d x_d) + \delta_{bc} (1 - Kx^d x_d) \right) \quad (258)$$

$$= Kx^a \left(\frac{K}{1 - Kx^\rho x_\rho} x_b x_c + \delta_{bc} \right). \quad (259)$$

Note that

$$\delta_{bc} = \partial_b x_c = \partial_b (C_{c\lambda} x^\lambda) = C_{c\lambda} \delta_b^\lambda = C_{cb}. \quad (260)$$

Then we can express the Christoffel symbols as:

$$\Gamma_{bc}^a = Kx^a \left(\frac{K}{1 - Kx^\rho x_\rho} x_b x_c + C_{bc} \right) = Kx^a g_{bc}. \quad (261)$$

(c)

The equations of motion for a geodesic are given by:

$$\frac{d^2 x^a}{d\lambda^2} + \Gamma_{bc}^a \frac{dx^b}{d\lambda} \frac{dx^c}{d\lambda} = 0, \quad (262)$$

where λ is the affine parameter. Substituting the expression for Γ_{bc}^a , we have:

$$\frac{d^2 x^a}{d\lambda^2} + Kx^a g_{bc} \frac{dx^b}{d\lambda} \frac{dx^c}{d\lambda} = 0. \quad (263)$$

This is a second-order ordinary differential equation for $x^a(\lambda)$. The solutions will depend on the value of K :

- For $K = 0$, the equation reduces to $\frac{d^2 x^a}{d\lambda^2} = 0$, which implies that $v^a = \frac{dx^a}{d\lambda}$ is constant. Thus, the solution is given by:

$$x^a(\lambda) = v^a \lambda + x_0^a, \quad (264)$$

where x_0^a is the initial position.

- For $K = +1$, the equation becomes $\frac{d^2 x^a}{d\lambda^2} + x^a g_{bc} \frac{dx^b}{d\lambda} \frac{dx^c}{d\lambda} = 0$. We can apply $\frac{d}{d\lambda}(g_{ab}v^a v^b)$ to find that $g_{ab}v^a v^b$ is a constant. Let $g_{ab}v^a v^b = \beta$, then we can express the equation as $\frac{d^2 x^a}{d\lambda^2} + \beta x^a = 0$. The solution to this equation is given by:

$$x^a(\lambda) = A^a \cos(\sqrt{\beta}\lambda) + B^a \sin(\sqrt{\beta}\lambda), \quad (265)$$

where A^a and B^a are constants determined by the initial conditions.

- For $K = -1$, the equation becomes $\frac{d^2 x^a}{d\lambda^2} - x^a g_{bc} \frac{dx^b}{d\lambda} \frac{dx^c}{d\lambda} = 0$. Similar to the case of $K = +1$, we can find that $g_{ab}v^a v^b$ is a constant. Let $g_{ab}v^a v^b = \beta$, then we can express the equation as $\frac{d^2 x^a}{d\lambda^2} - \beta x^a = 0$. The solution to this equation is given by:

$$x^a(\lambda) = C^a e^{\sqrt{\beta}\lambda} + D^a e^{-\sqrt{\beta}\lambda}, \quad (266)$$

where C^a and D^a are constants determined by the initial conditions.

□

Question 3

Problem 8.2 in Carroll. Note that in the 1st edition, there were two errors. As in class, the metric should contain e^{2Ht} instead of e^{Ht} . The problem should be solved for non comoving observers. That is, you can not take $dx^i/d\lambda = 0$.

Consider the de Sitter space in coordinates where the metric takes the form

$$ds^2 = dt^2 - e^{2Ht}(dx^2 + dy^2 + dz^2) \quad (267)$$

Solve the geodesic equation to find the affine parameter as a function of t . Show that geodesics reach $t \rightarrow -\infty$ in a finite affine parameter, demonstrating that these coordinates fail to cover the entire manifold.

Answer

The Christoffel symbols for the given metric can be computed as:

$$\Gamma_{\mu\nu}^{\lambda} = \frac{1}{2}g^{\lambda\sigma}(\partial_{\mu}g_{\nu\sigma} + \partial_{\nu}g_{\mu\sigma} - \partial_{\sigma}g_{\mu\nu}). \quad (268)$$

Since the metric is diagonal, we can compute the non-zero Christoffel symbols as:

$$\Gamma_{ij}^t = \frac{1}{2}g^{tt}(\partial_i g_{jt} + \partial_j g_{it} - \partial_t g_{ij}) \quad (269)$$

$$= \frac{1}{2}(1)(0 + 0 - \partial_t(-e^{2Ht}\delta_{ij})) \quad (270)$$

$$= He^{2Ht}\delta_{ij}, \quad (271)$$

and

$$\Gamma_{jt}^i = \frac{1}{2}g^{ik}(\partial_j g_{kt} + \partial_t g_{jk} - \partial_k g_{jt}) \quad (272)$$

$$= \frac{1}{2}(-e^{-2Ht}\delta^{ik})(0 + \partial_t(-e^{2Ht}\delta_{jk}) - 0) \quad (273)$$

$$= H\delta_j^i. \quad (274)$$

The geodesic equations are given by:

$$\frac{d^2x^{\lambda}}{d\lambda^2} + \Gamma_{\mu\nu}^{\lambda} \frac{dx^{\mu}}{d\lambda} \frac{dx^{\nu}}{d\lambda} = 0. \quad (275)$$

Substituting the expressions for the Christoffel symbols, we can write the geodesic equations as:

$$\frac{d^2t}{d\lambda^2} + He^{2Ht}\delta_{ij} \frac{dx^i}{d\lambda} \frac{dx^j}{d\lambda} = 0, \quad (276)$$

and

$$\frac{d^2 x^i}{d\lambda^2} + 2H \frac{dt}{d\lambda} \frac{dx^i}{d\lambda} = 0. \quad (277)$$

We can rewrite the second equation as:

$$\frac{d}{d\lambda} \left(e^{2Ht} \frac{dx^i}{d\lambda} \right) = 0 \quad (278)$$

$$\implies e^{2Ht} \frac{dx^i}{d\lambda} = C^i, \quad (279)$$

where C^i are constants determined by the initial conditions. Substituting this expression into the first equation, we have:

$$\frac{d^2 t}{d\lambda^2} + H e^{2Ht} \delta_{ij} \frac{C^i}{e^{2Ht}} \frac{C^j}{e^{2Ht}} = 0 \quad (280)$$

$$\implies \frac{d^2 t}{d\lambda^2} + H e^{-2Ht} \delta_{ij} C^i C^j = 0 \quad (281)$$

$$\implies \frac{d^2 t}{d\lambda^2} + H e^{-2Ht} C^2 = 0, \quad (282)$$

where $C^2 = \delta_{ij} C^i C^j$. This is a second-order ordinary differential equation for $t(\lambda)$. We consider the following equation:

$$\frac{d}{d\lambda} \left(\left(\frac{dt}{d\lambda} \right)^2 - e^{-2Ht} C^2 \right) \quad (283)$$

$$= 2 \frac{dt}{d\lambda} \frac{d^2 t}{d\lambda^2} + 2H e^{-2Ht} C^2 \frac{dt}{d\lambda} \quad (284)$$

$$= 2 \frac{dt}{d\lambda} \left(\frac{d^2 t}{d\lambda^2} + H e^{-2Ht} C^2 \right) \quad (285)$$

$$= 0. \quad (286)$$

Hence, we can find that $\left(\frac{dt}{d\lambda} \right)^2 - e^{-2Ht} C^2$ is a constant. Let $\left(\frac{dt}{d\lambda} \right)^2 - e^{-2Ht} C^2 = \beta$, then we can express the equation as:

$$\left(\frac{dt}{d\lambda} \right)^2 = \beta + e^{-2Ht} C^2 \quad (287)$$

$$\implies \frac{dt}{d\lambda} = \sqrt{\beta + e^{-2Ht} C^2}. \quad (288)$$

If $\beta = 0$, we can express the equation as:

$$\frac{dt}{d\lambda} = \sqrt{e^{-2Ht}C^2} \quad (289)$$

$$\implies \frac{dt}{d\lambda} = e^{-Ht}C \quad (290)$$

$$\implies \lambda = -\frac{1}{H}e^{Ht} + \lambda_0, \quad (291)$$

where λ_0 is a constant determined by the initial conditions. We can see that as $t \rightarrow -\infty$, $\lambda \rightarrow \lambda_0$, which means that geodesics reach $t \rightarrow -\infty$ in a finite affine parameter. This demonstrates that these coordinates fail to cover the entire manifold.

If $\beta \neq 0$, I don't know how to solve the equation analytically, but we can still find that as $t \rightarrow -\infty$, λ approaches a finite value, which also demonstrates that these coordinates fail to cover the entire manifold. $\square$

Problem Set 5 due on April 1 at 11:59pm

Question 1

In the Figure below

- (a) In the upper left corner of the figure, there are no solutions where the Universe is currently expanding from a singular state. Derive the analytic form for the boundary of the shaded region.
- (b) The curve above the line $\Omega_\Lambda = 0$ separates universes which either expand forever or recollapse. Derive the analytic form for this curve.
- (c) In the lower right, the shaded region corresponds to an exclusion due to the age of the Universe. Assuming $H_0 = 71$ km/s/Mpc calculate the age of the Universe along the boundary of the region.

Answer

(a)

We start with the Friedmann Equation:

$$H^2 = H_0^2(\Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda), \quad (292)$$

where H is the Hubble parameter, H_0 is the current Hubble parameter, Ω_m is the matter (dark plus baryonic) density, Ω_k is the spatial curvature density, Ω_Λ is the vacuum density. If we define $x = 1/a$, and try to find the roots for the equation $f(x) = (H/H_0)^2 = 0$, it's given by:

$$f(x) = \Omega_m x^3 + \Omega_k x^2 + \Omega_\Lambda = 0 \quad (293)$$

$$\implies \Omega_m x^3 + (1 - \Omega_m - \Omega_\Lambda)x^2 + \Omega_\Lambda = 0, \quad \text{by } \Omega_k = 1 - \Omega_m - \Omega_\Lambda. \quad (294)$$

If we need the "no big bang" situation, we need $f'(x_0) = 0$, where x_0 is the root. In other words, $f(x)$ has a double root. Then, we have

$$f'(x) = 3\Omega_m x^2 + 2(1 - \Omega_m - \Omega_\Lambda)x = 0, \quad (295)$$

and the solution x_0 is given by

$$x_0 = \frac{2(\Omega_m + \Omega_\Lambda - 1)}{3\Omega_m}. \quad (296)$$

Now we require $f(x_0) = 0$:

$$\Omega_m \left(\frac{2(1 - \Omega_m - \Omega_\Lambda)}{3\Omega_m} \right)^3 - (1 - \Omega_m - \Omega_\Lambda) \left(\frac{2(1 - \Omega_m - \Omega_\Lambda)}{3\Omega_m} \right)^2 + \Omega_\Lambda = 0 \quad (297)$$

$$\Rightarrow \Omega_m \left(\frac{2\alpha}{3\Omega_m} \right)^3 - \alpha \left(\frac{2\alpha}{3\Omega_m} \right)^2 + \Omega_\Lambda = 0, \quad \alpha = 1 - \Omega_m - \Omega_\Lambda \quad (298)$$

$$\Rightarrow \frac{8\alpha^3}{27\Omega_m^2} - \frac{4\alpha^3}{9\Omega_m^2} = -\Omega_\Lambda, \quad (299)$$

$$\Rightarrow \Omega_\Lambda = \frac{4(\Omega_m + \Omega_\Lambda - 1)^3}{27\Omega_m^2} \quad (300)$$

$$\Rightarrow 3\Omega_m^{2/3}\Omega_\Lambda^{1/3} = 2^{2/3}(\Omega_m + \Omega_\Lambda - 1). \quad (301)$$

We define $\Omega_\Lambda = 4\Omega_m \cos^3 \theta$. Then we have

$$3\Omega_m \cos \theta = 4\Omega_m \cos^3 \theta + \Omega_m - 1 \quad (302)$$

$$\Rightarrow \frac{1 - \Omega_m}{\Omega_m} = 4 \cos^3 \theta - 3 \cos \theta = \cos 3\theta \quad (303)$$

$$\Rightarrow \theta = \frac{1}{3} \arccos \left(\frac{1 - \Omega_m}{\Omega_m} \right) \quad (304)$$

$$\Rightarrow \Omega_\Lambda = 4\Omega_m \left[\cos \left(\frac{1}{3} \arccos \left(\frac{1 - \Omega_m}{\Omega_m} \right) \right) \right]^3. \quad (305)$$

Note that this solution is valid provided $\Omega_m \geq 0.5$. If the $\Omega_m \leq 0.5$, we can use hyperbolic cosine function to parametrize the equation, and apply the same procedure. Then we can get

$$\Omega_\Lambda = 4\Omega_m \left[\cosh \left(\frac{1}{3} \operatorname{arccosh} \left(\frac{1 - \Omega_m}{\Omega_m} \right) \right) \right]^3. \quad (306)$$

(b)

We start with the same cubic relation derived from the double root condition:

$$\cos(3\theta) = \frac{1 - \Omega_m}{\Omega_m} \quad (307)$$

The general solution for 3θ is given by:

$$3\theta = \arccos \left(\frac{1 - \Omega_m}{\Omega_m} \right) + 2n\pi, \quad n = 0, 1, 2 \quad (308)$$

To identify the curve that separates "eternal expansion" from "recollapse," we require the boundary to pass through the specific point $(\Omega_m, \Omega_\Lambda) = (1, 0)$. Testing the roots at $\Omega_m = 1$ (where $\arccos(0) = \pi/2$):

- For $n = 0$: $\theta = \pi/6$, leading to $\Omega_\Lambda \approx 2.6$ (This corresponds to the No Big Bang boundary).
- For $n = 1$: $\theta = 5\pi/6$, leading to $\Omega_\Lambda < 0$.

- For $n = 2$: $\theta = \frac{\pi/2+4\pi}{3} = \frac{3\pi}{2}$, which gives $\cos(3\pi/2) = 0$, and thus $\Omega_\Lambda = 0$.

Therefore, we choose the branch $n = 2$. The analytic form for this curve is:

$$\Omega_\Lambda = 4\Omega_m \left[\cos \left(\frac{1}{3} \arccos \left(\frac{1 - \Omega_m}{\Omega_m} \right) + \frac{4\pi}{3} \right) \right]^3 \quad (309)$$

(c)

We start the equation:

$$t_0 = \int_0^{t_0} dt = \int_0^1 \frac{da}{aH(a)} \quad (310)$$

$$= \frac{1}{H_0} \int_0^1 \frac{da}{\sqrt{\Omega_m a^{-1} + \Omega_k + \Omega_\Lambda a^2}}. \quad (311)$$

We consider the three cases $(\Omega_m, \Omega_\Lambda, \Omega_k) = (2, -1.5, 0.5)$, $(3, -0.5, -1.5)$ and $(3, -1.5, -0.5)$, then we get:

$$t_0 = \frac{0.504}{H_0} = 6.89 \text{ Gyr}, \quad (312)$$

$$t_0 = \frac{0.495}{H_0} = 6.76 \text{ Gyr}, \quad (313)$$

and

$$t_0 = \frac{0.463}{H_0} = 6.33 \text{ Gyr}. \quad (314)$$

These results are significantly lower than the estimated ages of the oldest stars and globular clusters (> 12 Gyr). Thus, the lower-right region of the $\Omega_m - \Omega_\Lambda$ plane is excluded as it leads to a "young universe" paradox. $\square$

Question 2

Find the proper distance and determine the time the light was emitted

- (a) to a galaxy with redshift $z = 0.01$
- (b) to a galaxy with redshift $z = 3$
- (c) What is the angle spanned by the horizon at a redshift $z = 1100$? You will need to know the size of the horizon at that redshift and the (angular) distance to that redshift.

We will assume matter ($\gamma = 1$) and compare the results for (i) $(\Omega_0, \Omega_\Lambda) = (1, 0)$ and (ii) $(\Omega_0, \Omega_\Lambda) = (0.27, 0.73)$.

Answer

Now, we start with the equations of proper distance and time of emission:

$$d_p = c \int_0^z \frac{dz}{H}, \quad (315)$$

$$t_e = \int_z^\infty \frac{dz}{(1+z)H}, \quad (316)$$

where z is red shifting, $H = H_0 \sqrt{\Omega_0 a^{-3} + \Omega_\Lambda}$, $a^{-1} = 1 + z$ Then we can have

$$d_p = \frac{c}{H_0} \int_0^z \frac{dz}{\sqrt{\Omega_0(1+z)^3 + \Omega_\Lambda}}, \quad (317)$$

$$t_e = \frac{1}{H_0} \int_z^\infty \frac{dz}{(1+z)\sqrt{\Omega_0(1+z)^3 + \Omega_\Lambda}} \quad (318)$$

(i) If $(\Omega_0, \Omega_\Lambda) = (1, 0)$,

(a) $(d_p, t_e) = (41.9357 \text{ Mpc}, 8.97175 \text{ Gyr})$

(b) $(d_p, t_e) = (4225 \text{ Mpc}, 1.1383 \text{ Gyr})$

(ii) If $(\Omega_0, \Omega_\Lambda) = (0.27, 0.73)$

(a) $(d_p, t_e) = (42.1642 \text{ Mpc}, 13.4245 \text{ Gyr})$

(b) $(d_p, t_e) = (6465.85 \text{ Mpc}, 2.1755 \text{ Gyr})$

(c)

The angular size θ subtended by the horizon is defined as the ratio of the physical horizon size at redshift z to the angular diameter distance d_a :

$$\theta = \frac{d_{H,phys}(z)}{d_a(z)} = \frac{r_H/(1+z)}{d_p/(1+z)} = \frac{r_H}{d_p}. \quad (319)$$

Since our models are flat ($\Omega_k = 0$), the factor $(1+z)$ cancels out, and the angle can be directly computed using comoving distances. Now for the angle spanned θ , we have

$$\theta = \frac{r_H(z)}{d_p(z)} \quad (320)$$

where

$$r_H = c \int_z^\infty \frac{dz}{H} = \frac{c}{H_0} \int_z^\infty \frac{dz}{\sqrt{\Omega_0(1+z)^3 + \Omega_\Lambda}}. \quad (321)$$

Then we have

- (i) If $(\Omega_0, \Omega_\Lambda) = (1, 0)$, $\theta = \frac{254.66}{8195} = 1.78^\circ$
- (ii) If $(\Omega_0, \Omega_\Lambda) = (0.27, 0.73)$, $\theta = \frac{490.09}{14082} = 1.99^\circ$

□

Question 3

A particle χ with mass $m = 1$ GeV and $g_\chi = 2$ as a cross-section for the process $\chi\bar{\chi} \rightarrow e^+e^-$ given by

$$\langle\sigma v\rangle = \frac{T^2}{(T^2 + m^2)^2}. \quad (322)$$

Between what temperatures is χ in thermal equilibrium with the electron?

Answer

The density n_χ is given by

$$n_\chi = \frac{g_\chi}{(2\pi)^3} \int f(\mathbf{p}) d^3p = \frac{g_\chi}{2\pi^2} \int_0^\infty \frac{p^2 dp}{\exp(\sqrt{p^2 + m^2}/T) \pm 1} \quad (323)$$

If we assume the $T \ll m$, we can have

$$n_\chi \approx \frac{g_\chi}{2\pi^2} \int_0^\infty \frac{p^2 dp}{\exp(\sqrt{p^2 + m^2}/T)} \quad (324)$$

$$= \frac{g_\chi}{2\pi^2} \int_0^\infty p^2 \exp(-\sqrt{p^2 + m^2}/T) dp \quad (325)$$

$$\approx \frac{g_\chi}{2\pi^2} \int_0^\infty p^2 \exp\left[-\frac{1}{T} \left(m + \frac{p^2}{2m}\right)\right], \quad E = \sqrt{p^2 + m^2} \approx m + \frac{p^2}{2m} \quad (326)$$

$$= \frac{g_\chi}{2\pi^2} \sqrt{\frac{\pi}{2}} (mT)^{3/2} e^{-m/T} \quad (327)$$

$$= g_\chi \left(\frac{mT}{2\pi}\right)^{3/2} e^{-m/T} = 2 \left(\frac{mT}{2\pi}\right)^{3/2} e^{-m/T}. \quad (328)$$

Now, we can calculate the interaction rate Γ , given by

$$\Gamma = \langle\sigma v\rangle n = \frac{T^2}{(T^2 + m^2)^2} 2 \left(\frac{mT}{2\pi}\right)^{3/2} e^{-m/T}. \quad (329)$$

The Hubble parameter is given by

$$H(T) = \sqrt{\frac{4\pi^3}{45}} g \frac{T^2}{M_{Pl}} \approx 1.66 \frac{\sqrt{g} T^2}{M_{Pl}}, \quad (330)$$

where the Planck mass M_{Pl} is 1.22×10^{19} GeV. Now we can try to solve the root for the equation $\Gamma(T) = H(T)$, given by

$$\frac{T^2}{(T^2 + m^2)^2} 2 \left(\frac{mT}{2\pi} \right)^{3/2} e^{-m/T} = 1.66 \frac{\sqrt{g} T^2}{M_{Pl}} \quad (331)$$

$$\Rightarrow \frac{2}{(T^2 + 1)^2} \left(\frac{T}{2\pi} \right)^{3/2} e^{-1/T} = 1.66 \frac{\sqrt{g}}{1.22 \times 10^{19}}, \quad \text{substituting } m = 1 \text{ GeV} \quad (332)$$

To find the temperature range where χ is in thermal equilibrium with electrons, we solve $\Gamma(T) \geq H(T)$. Based on the numerical solution of the transcendental equation:

$$\frac{2}{(T^2 + 1)^2} \left(\frac{T}{2\pi} \right)^{3/2} e^{-1/T} = \frac{1.66\sqrt{100}}{1.22 \times 10^{19}}, \quad (333)$$

where we simply assume $g \approx 100$. Then we can obtain two solutions

$$T_{\text{low}} = 0.0295 \text{ GeV} = 29.5 \text{ MeV}, \quad (334)$$

$$T_{\text{high}} = 6.13 \times 10^6 \text{ GeV}. \quad (335)$$

Between 29.5 MeV and 6×10^6 GeV, the particle χ can be in thermal equilibrium with the electrons. Note that we have to consider different g in the low temperature. When the temperature is below $\mathcal{O}(10)$ MeV, we consider photons γ , neutrinos ν , and electron e . Hence, the g is given by

$$g = 2 + 3 \times 2 \times \frac{7}{8} + 3 \times 2 \times \frac{7}{8} = 10.75. \quad (336)$$

With this new g , the new answer for T_{low} become

$$T_{\text{low}} = 0.0286 \text{ GeV} = 28.6 \text{ MeV} \quad (337)$$

Hence, between 28.6 MeV and 6×10^6 GeV, the particle χ can be in thermal equilibrium with the electrons. **However, this statement is wrong again.** With this T_{high} , χ is no longer non-relativistic. We have to re-derive this relation again for relativistic case. Start with number density n_χ given by

$$n_\chi = \frac{g_\chi}{(2\pi)^3} \int f(\mathbf{p}) d^3p = \frac{g_\chi}{2\pi^2} \int_0^\infty \frac{p^2 dp}{\exp(\sqrt{p^2 + m^2}/T) \pm 1} \quad (338)$$

$$\approx \frac{g_\chi}{2\pi^2} \int_0^\infty \frac{p^2 dp}{\exp(p/T) \pm 1} \quad (339)$$

$$= \begin{cases} \frac{g_\chi}{2\pi^2} \frac{3\zeta(3)}{2} T^3, & \text{boson} \\ \frac{g_\chi}{2\pi^2} 2\zeta(3) T^3, & \text{fermion} \end{cases}. \quad (340)$$

Now we can do the similar process again. The interaction rate is given by

$$\Gamma = \begin{cases} \frac{T^2}{(T^2+m^2)^2} \frac{g_\chi}{2\pi^2} \frac{3\zeta(3)}{2} T^3, & \text{boson} \\ \frac{T^2}{(T^2+m^2)^2} \frac{g_\chi}{2\pi^2} 2\zeta(3) T^3, & \text{fermion} \end{cases} \quad (341)$$

$$= \begin{cases} \frac{T^2}{(T^2+m^2)^2} \frac{3\zeta(3)}{2\pi^2} T^3, & \text{boson} \\ \frac{T^2}{(T^2+m^2)^2} \frac{4\zeta(3)}{2\pi^2} T^3, & \text{fermion} \end{cases} \quad (342)$$

Compare with hubble parameter, we can obtain

$$\begin{cases} \frac{T^2}{(T^2+1)^2} \frac{3\zeta(3)}{2\pi^2} T^3 = 1.66 \frac{\sqrt{g} T^2}{M_{Pl}}, & \text{boson} \\ \frac{T^2}{(T^2+1)^2} \frac{4\zeta(3)}{2\pi^2} T^3 = 1.66 \frac{\sqrt{g} T^2}{M_{Pl}}, & \text{fermion} \end{cases} \quad (343)$$

$$\Rightarrow \begin{cases} \frac{1}{(T^2+1)^2} \frac{3\zeta(3)}{2\pi^2} T^3 = 1.66 \frac{\sqrt{100}}{M_{Pl}}, & \text{boson} \\ \frac{1}{(T^2+1)^2} \frac{4\zeta(3)}{2\pi^2} T^3 = 1.66 \frac{\sqrt{100}}{M_{Pl}}, & \text{fermion} \end{cases} \quad (344)$$

$$\Rightarrow \begin{cases} T_{\text{high}} = 1.34267 \times 10^{17} \text{ GeV}, & (\text{boson}) \\ T_{\text{high}} = 1.79022 \times 10^{17} \text{ GeV}, & (\text{fermion}) \end{cases} \quad (345)$$

Here, we consider the SM to estimate $g \approx 100$. In summary, by considering the appropriate relativistic limits and effective degrees of freedom, we find that the particle χ is in thermal equilibrium with the thermal bath in the temperature range:

$$28.6 \text{ MeV} \lesssim T \lesssim 1.79 \times 10^{17} \text{ GeV} \quad (\text{assuming fermion } \chi) \quad (346)$$

The lower bound is determined by the non-relativistic freeze-out, while the upper bound represents the initial coupling of the species in the very early universe. $\square$

Problem Set 6 due on April 15 at 11:59pm

Question 1

Find the dependence of ΔY on the number of neutrino flavors, N_ν . That is, find the ratio $(\Delta Y/Y)/(\Delta N_\nu/N_\nu)$. Evaluate this. Remember that there are two sources to consider. The effect on n/p freeze-out and the effect on time at which BBN starts.

Answer

First, the Y is given by

$$Y = 2 \frac{n}{p+n} = \frac{2n/p}{1+n/p} = \frac{2x}{1+x} \quad (347)$$

$$\Rightarrow \frac{dY}{dx} = 2 \frac{(1+x) - x}{(1+x)^2} = \frac{2}{(1+x)^2} \quad (348)$$

$$\Rightarrow \Delta Y = \frac{2\Delta x}{(1+x)^2} \quad (349)$$

$$\Rightarrow \frac{\Delta Y}{Y} = \left(\frac{1+x}{2x} \right) \frac{2}{(1+x)^2} \Delta x = \frac{1}{x(1+x)} \Delta x \quad (350)$$

$$= \frac{1}{1+x} \frac{\Delta x}{x} \quad (351)$$

The ratio x can be determined by the mass difference between protons and neutrons $\Delta m = m_n - m_p$, which is given by:

$$x = x_f e^{-t_{\text{BBN}}/\tau_n}, \quad (352)$$

where $x_f = e^{-\Delta m/T_f}$, τ_n is the decay constant of the neutron. To find $\frac{\Delta x}{x}$, we take the logarithm of x and vary it:

$$\ln x = -\frac{\Delta m}{T_f} - \frac{t_{\text{BBN}}}{\tau_n} \quad (353)$$

$$\Rightarrow \frac{\Delta x}{x} = \left(\frac{\Delta m}{T_f} \right) \frac{\Delta T_f}{T_f} - \frac{\Delta t_{\text{BBN}}}{\tau_n} \quad (354)$$

Now we consider the two sources as requested:

1. Effect on n/p freeze-out (T_f): The freeze-out occurs when the weak interaction rate $\Gamma_w \approx G_F^2 T^5$ falls below the Hubble expansion rate $H \approx \sqrt{g_*} T^2 / M_{Pl}$. This implies $T_f \propto g_*^{1/6}$. The effective

degrees of freedom g_* depends on N_ν as $g_* = \frac{11}{2} + \frac{7}{4}N_\nu$. Therefore:

$$\frac{\Delta T_f}{T_f} = \frac{1}{6} \frac{\Delta g_*}{g_*} = \frac{1}{6} \left(\frac{7/4}{g_*} \right) \Delta N_\nu \quad (355)$$

Substituting this into the first term:

$$\left(\frac{\Delta x}{x} \right)_{\text{freeze-out}} = \frac{\Delta m}{6T_f} \frac{\Delta g}{g_*} \quad (356)$$

2. Effect on the time BBN starts (t_{BBN}): BBN begins at a fixed temperature $T_{\text{BBN}} \approx 0.07$ MeV. In the radiation-dominated era, the time-temperature relation is $t \propto g_*^{-1/2} T^{-2}$. Thus, $t_{\text{BBN}} \propto g_*^{-1/2}$, leading to:

$$\frac{\Delta t_{\text{BBN}}}{t_{\text{BBN}}} = -\frac{1}{2} \frac{\Delta g_*}{g_*} \quad (357)$$

$$\Rightarrow -\frac{\Delta t_{\text{BBN}}}{\tau_n} = -\frac{t_{\text{BBN}}}{\tau_n} \left(-\frac{1}{2} \frac{\Delta g_*}{g_*} \right) = \frac{t_{\text{BBN}}}{2\tau_n} \frac{\Delta g_*}{g_*} \quad (358)$$

Combining both effects:

$$\frac{\Delta x}{x} = \left(\frac{\Delta m}{6T_f} + \frac{t_{\text{BBN}}}{2\tau_n} \right) \frac{\Delta g_*}{g_*} \quad (359)$$

Now we evaluate the sensitivity ratio $(\Delta Y/Y)/(\Delta N_\nu/N_\nu)$. Using $g_* \approx 10.75$ and $\Delta g_* = \frac{7}{4} \Delta N_\nu$:

$$\frac{\Delta Y/Y}{\Delta N_\nu/N_\nu} = \left(\frac{1}{1+x} \right) \left(\frac{\Delta m}{6T_f} + \frac{t_{\text{BBN}}}{2\tau_n} \right) \frac{7N_\nu}{4g_*} \quad (360)$$

To evaluate the ratio, we use the standard values at the BBN epoch:

- Neutron-to-proton ratio: $x \approx 1/7$
- Mass difference: $\Delta m \approx 1.29$ MeV
- Freeze-out temperature: $T_f \approx 0.8$ MeV
- BBN start time: $t_{\text{BBN}} \approx 200$ s
- Neutron lifetime: $\tau_n \approx 880$ s
- Effective degrees of freedom: $g_* = 10.75$ (for $N_\nu = 3$)

Finally, we get

$$\frac{\Delta Y/Y}{\Delta N_\nu/N_\nu} \approx \mathbf{0.164} \quad (361)$$

Question 2

A new interaction is discovered as well as three new neutrinos which interact only through this new interaction. If as in the case of the weak interactions, this new rate is proportional to some $M_{Z'}^{-4}$, find the cosmological limit on the mass of Z' . To do this, assume that the limit from BBN on the number of neutrinos is $N_\nu < 3.2$.

Answer

First, the interaction rate can be determined by

$$\Gamma = \frac{T^5}{M_{Z'}^4}. \quad (362)$$

We can take T^5 to match the dimension on both sides. Decoupling occurs at temperature T_d when $\Gamma(T_d) \approx H(T_d)$.

$$\frac{T_d^5}{M_{Z'}^4} \approx \frac{T_d^2}{M_{Pl}} \quad (363)$$

$$\implies T_d^3 \approx \frac{M_{Z'}^4}{M_{Pl}} \quad (364)$$

where $M_{Pl} \approx 10^{19}$ GeV is the Planck mass. After the new neutrinos decouple at T_d , the entropy of the rest of the Standard Model (SM) plasma is conserved. As other SM particles annihilate, they heat the photons and standard neutrinos, but not the decoupled ν_s . Using entropy conservation $s \propto g_* T^3 = \text{const}$:

$$\frac{T_{\nu s}}{T_\nu} = \left(\frac{g(T_{BBN})}{g(T_d)} \right)^{1/3} \quad (365)$$

where $g_*(T_{BBN}) = 10.75$. The contribution of the 3 new neutrino flavors to the effective number of neutrinos is:

$$\Delta N_\nu = 3 \times \left(\frac{T_{\nu s}}{T_\nu} \right)^4 = 3 \times \left(\frac{10.75}{g_*(T_d)} \right)^{4/3} \quad (366)$$

Given the BBN limit $N_\nu < 3.2$, we have $\Delta N_\nu < 0.2$. Solving for $g_*(T_d)$:

$$3 \times \left(\frac{10.75}{g(T_d)} \right)^{4/3} < 0.2 \quad (367)$$

$$\left(\frac{10.75}{g(T_d)} \right)^{4/3} < 0.0667 \quad (368)$$

$$\implies g_*(T_d) > \frac{10.75}{(0.0667)^{3/4}} \approx 82.2 \quad (369)$$

In the SM, $g_* > 82$ corresponds to a temperature T_d above the QCD phase transition ($\Lambda_{QCD} \approx 200$ MeV). At $T \approx 10$ GeV (where all quarks except top are relativistic), $g_* \approx 86$. Let's assume $T_d \approx 10$ GeV. Finally, substituting T_d back into Eq. (1):

$$M_{Z'}^4 \approx T_d^3 \times M_{Pl} \tag{370}$$

$$M_{Z'}^4 \approx (10 \text{ GeV})^3 \times 10^{19} \text{ GeV} = 10^{22} \text{ GeV}^4 \tag{371}$$

$$\implies M_{Z'} \approx 10^{5.5} \text{ GeV} = 3.16 \times 10^5 \text{ GeV} = 316 \text{ TeV} \tag{372}$$

□

Question 3

For a neutrino mass of $m_\nu = 10$ GeV, determine the ratio T_ν/T_γ needed to compute the relic density of neutrinos. What is the temperature ratio for $m_\nu = 100$ GeV?

Answer

The contribution of a massive neutrino species to the energy density of the universe today is given by the formula (c.f. eq (3.2.79) in Baumann's note):

$$\Omega_\nu h^2 = \frac{\sum m_\nu}{94 \text{ eV}} \left(\frac{T_\nu}{T_\gamma} \right)^3 \quad (373)$$

In the standard case where $T_\nu/T_\gamma = (4/11)^{1/3} \approx 0.714$, we recover the standard result for light neutrinos. However, for very heavy neutrinos, we must ensure $\Omega_\nu h^2 \leq \Omega_c h^2 \approx 0.12$ to avoid overclosing the universe. Setting the relic density to the observed dark matter density $\Omega_\nu h^2 \approx 0.12$, we solve for the ratio $(T_\nu/T_\gamma)^3$:

$$0.12 = \frac{10 \times 10^9 \text{ eV}}{94 \text{ eV}} \left(\frac{T_\nu}{T_\gamma} \right)^3 \quad (374)$$

$$\left(\frac{T_\nu}{T_\gamma} \right)^3 = \frac{0.12 \times 94}{10^{10}} \approx 1.128 \times 10^{-9} \quad (375)$$

$$\Rightarrow \frac{T_\nu}{T_\gamma} = (1.128 \times 10^{-9})^{1/3} \approx \mathbf{1.04 \times 10^{-3}} \quad (376)$$

Similarly, for a 100 GeV neutrino:

$$0.12 = \frac{100 \times 10^9 \text{ eV}}{94 \text{ eV}} \left(\frac{T_\nu}{T_\gamma} \right)^3 \quad (377)$$

$$\left(\frac{T_\nu}{T_\gamma} \right)^3 = \frac{0.12 \times 94}{10^{11}} \approx 1.128 \times 10^{-10} \quad (378)$$

$$\Rightarrow \frac{T_\nu}{T_\gamma} = (1.128 \times 10^{-10})^{1/3} \approx \mathbf{4.82 \times 10^{-4}} \quad (379)$$

□

Problem Set 7 due on April 27 at 11:59pm

Question 1

In a baryon symmetric universe, the final baryon (and anti-baryon) density is roughly $n_B/n_\gamma = 10^{-18}$. In a baryon asymmetric universe with $\eta = (n_B - n_{\bar{B}})/n_\gamma \approx 6 \times 10^{-10}$, determine the freeze-out temperature for annihilations and the final density of anti-baryons. Note the freeze out temperature for baryons is NOT the same as for antibaryons (if there is an asymmetry).

Answer

Assuming a radiation-dominated universe, we can express the Hubble parameter as:

$$H = \sqrt{\frac{8\pi G\rho}{3}} = \sqrt{\frac{8\pi Gg_*\pi^2 T^4}{90}}, \quad (380)$$

where g_* is the effective number of relativistic degrees of freedom. The baryon number density can be expressed in terms of the baryon-to-photon ratio η and the photon number density n_γ :

$$n_B \approx \eta n_\gamma = \eta \frac{2\zeta(3)}{\pi^2} T^3 \quad (381)$$

The freeze-out condition for anti-baryons is defined when the annihilation rate $\Gamma_{\bar{B}}$ equals the Hubble expansion rate H :

$$\Gamma_{\bar{B}} = n_B \langle \sigma v \rangle = H \quad (382)$$

Substituting the expressions for n_B and H into the freeze-out condition:

$$\left(\eta \frac{2\zeta(3)}{\pi^2} T^3 \right) \langle \sigma v \rangle = \sqrt{\frac{8\pi^3 Gg_*}{90}} T^2 \quad (383)$$

Solving for the anti-baryon freeze-out temperature $T_{\bar{f}}$:

$$T_{\bar{f}} = \frac{\pi^2}{\eta \cdot 2\zeta(3) \cdot \langle \sigma v \rangle} \sqrt{\frac{8\pi^3 Gg_*}{90}} \quad (384)$$

Note: Since η is in the denominator, a larger asymmetry results in a lower freeze-out temperature, meaning anti-baryons continue to annihilate for a longer period.

The presence of the baryon excess η exerts an exponential suppression on the anti-baryon abundance. The final ratio of anti-baryons to photons ($n_{\bar{B}}/n_\gamma$) can be derived from the integrated Boltzmann

equation for an asymmetric species:

$$\frac{n_{\bar{B}}}{n_{\gamma}} \approx \eta \exp \left(-\frac{\eta}{(n_B/n_{\gamma})_{sym}} \right) \quad (385)$$

where $(n_B/n_{\gamma})_{sym} \approx 10^{-18}$ is the residual density in a symmetric universe.

Using the observed values:

- $\eta \approx 6 \times 10^{-10}$
- $(n_B/n_{\gamma})_{sym} \approx 10^{-18}$

The calculation yields:

$$\frac{n_{\bar{B}}}{n_{\gamma}} \approx (6 \times 10^{-10}) \cdot \exp \left(-\frac{6 \times 10^{-10}}{10^{-18}} \right) = (6 \times 10^{-10}) \cdot e^{-6 \times 10^8} \quad (386)$$

The factor $e^{-6 \times 10^8}$ is vanishingly small, effectively zero in any physical context. Thus, the final density of anti-baryons is negligible:

$$\frac{n_{\bar{B}}}{n_{\gamma}} \approx 0 \quad (387)$$

In conclusion, the freeze-out temperature for anti-baryons is significantly lower than that for baryons due to the asymmetry, and the final density of anti-baryons is effectively zero in a baryon asymmetric universe.

Question 2

Analogous to the calculation of Q in terms of chemical potentials done in class (see notes), Calculate expressions for Q_3 , B and L that is weak isospin, Baryon number and lepton number. From these results setting $Q = Q_3 = 0$, and $9\mu_{u_L} + 3\mu_\nu = 0$ for sphaleron interactions in equilibrium, find the ratio $B/(B - L)$.

Answer

In the high-temperature limit ($T \gg m$), the difference between the number density of a particle species and its antiparticle is given by:

$$n_i - n_{\bar{i}} \approx \frac{g_i T^2}{6} \mu_i \quad (\text{for fermions}), \quad \frac{g_i T^2}{3} \mu_i \quad (\text{for bosons}) \quad (388)$$

We assume N generations of fermions ($N = 3$). Let $\mu_{u_L}, \mu_{d_L}, \mu_{u_R}, \mu_{d_R}$ be quark potentials, $\mu_\nu, \mu_{e_L}, \mu_{e_R}$ be lepton potentials, and μ_W, μ_ϕ be the potentials for W bosons and Higgs scalars. First, for the weak isospin Q_3 , we sum over all left-handed fermions and bosons:

$$Q_3 = 3 \cdot \frac{T^2}{6} \left[3 \cdot \frac{1}{2} (\mu_{u_L} - \mu_{d_L}) + \frac{1}{2} (\mu_\nu - \mu_{e_L}) \right] + 2 \cdot \frac{T^2}{3} \mu_W \quad (389)$$

$$= \frac{T^2}{6} \left[\frac{9}{2} (\mu_{u_L} - \mu_{d_L}) + \frac{3}{2} (\mu_\nu - \mu_{e_L}) + 4\mu_W \right] \quad (390)$$

The factor of 3 accounts for the color degrees of freedom for quarks, and the factor of 2 accounts for the two polarization states of the W bosons. Next, for the baryon number B , we sum over all quarks:

$$B = 3 \cdot 3 \cdot \frac{T^2}{6} \left[\frac{1}{3} (\mu_{u_L} + \mu_{d_L}) + \frac{1}{3} (\mu_{u_R} + \mu_{d_R}) \right] \quad (391)$$

$$= \frac{T^2}{6} [3(\mu_{u_L} + \mu_{d_L}) + 3(\mu_{u_R} + \mu_{d_R})] \quad (392)$$

For the lepton number L , we sum over all leptons:

$$L = N \cdot \frac{T^2}{6} [\mu_\nu + \mu_{e_L} + \mu_{e_R}] \quad (393)$$

$$= \frac{T^2}{6} [3\mu_\nu + 3\mu_{e_L} + 3\mu_{e_R}]. \quad (394)$$

The Q is given by the sum of the electric charges of all particles:

$$Q = 3 \cdot 3 \cdot \frac{T^2}{6} \left[\frac{2}{3}(\mu_{u_L} + \mu_{u_R}) - \frac{1}{3}(\mu_{d_L} + \mu_{d_R}) \right] + 3 \cdot \frac{T^2}{6} [-(\mu_{e_L} + \mu_{e_R})] + 2 \cdot \frac{T^2}{3} \mu_W + 2 \cdot \frac{T^2}{3} \mu_\phi \quad (395)$$

$$= \frac{T^2}{6} \left[3 \cdot 3 \cdot \left(\frac{2}{3}(\mu_{u_L} + \mu_{u_R}) - \frac{1}{3}(\mu_{d_L} + \mu_{d_R}) \right) - 3 \cdot (\mu_{e_L} + \mu_{e_R}) + 4\mu_W + 4\mu_\phi \right] \quad (396)$$

$$= \frac{T^2}{6} [6(\mu_{u_L} + \mu_{u_R}) - 3(\mu_{d_L} + \mu_{d_R}) - 3(\mu_{e_L} + \mu_{e_R}) + 4\mu_W + 4\mu_\phi]. \quad (397)$$

Now, we set $Q = 0$ and $Q_3 = 0$ to find relations between the chemical potentials. From $Q_3 = 0$, we have:

$$\frac{9}{2}(\mu_{u_L} - \mu_{d_L}) + \frac{3}{2}(\mu_\nu - \mu_{e_L}) + 4\mu_W = 0 \quad \Rightarrow \quad 9(\mu_{u_L} - \mu_{d_L}) + 3(\mu_\nu - \mu_{e_L}) + 8\mu_W = 0 \quad (398)$$

From $Q = 0$, we have:

$$6(\mu_{u_L} + \mu_{u_R}) - 3(\mu_{d_L} + \mu_{d_R}) - 3(\mu_{e_L} + \mu_{e_R}) + 4\mu_W + 4\mu_\phi = 0 \quad (399)$$

Next, the sphaleron, yukawa interactions, and weak interactions in equilibrium give us the following relations:

$$\mu_\nu = -3\mu_{u_L} \quad (400)$$

$$\mu_{e_R} = \mu_{e_L} - \mu_\phi \quad (401)$$

$$\mu_{u_R} = \mu_{u_L} + \mu_\phi \quad (402)$$

$$\mu_{d_R} = \mu_{d_L} - \mu_\phi \quad (403)$$

$$\mu_{d_L} = \mu_{u_L} + \mu_W \quad (404)$$

$$\mu_{e_L} = \mu_\nu + \mu_W \quad (405)$$

Substituting these relations into the $Q = 0$ equation, we get:

$$6(\mu_{u_L} + \mu_{u_L} + \mu_\phi) - 3(\mu_{u_L} + \mu_W + \mu_{u_L} + \mu_W - \mu_\phi) - 3(\mu_\nu + \mu_W + \mu_{e_L} - \mu_\phi) + 4\mu_W + 4\mu_\phi = 0 \quad (406)$$

$$\rightarrow 12\mu_{u_L} + 6\mu_\phi - 6\mu_{u_L} - 6\mu_W + 3\mu_\phi - 3\mu_\nu - 3\mu_W - 3\mu_{e_L} + 3\mu_\phi + 4\mu_W + 4\mu_\phi = 0 \quad (407)$$

$$\rightarrow 6\mu_{u_L} + 16\mu_\phi - 5\mu_W - 3\mu_\nu - 3\mu_{e_L} = 0 \quad (408)$$

Substituting $\mu_\nu = -3\mu_{u_L}$ and $\mu_{e_L} = \mu_\nu + \mu_W$ into the above equation, we get:

$$6\mu_{u_L} + 16\mu_\phi - 5\mu_W - 3(-3\mu_{u_L}) - 3(-3\mu_{u_L} + \mu_W) = 0 \quad (409)$$

$$\rightarrow 6\mu_{u_L} + 16\mu_\phi - 5\mu_W + 9\mu_{u_L} + 9\mu_{u_L} - 3\mu_W = 0 \quad (410)$$

$$\rightarrow 24\mu_{u_L} + 16\mu_\phi - 8\mu_W = 0 \quad (411)$$

Substituting $\mu_{d_L} = \mu_{u_L} + \mu_W$ into the $Q_3 = 0$ equation, we get:

$$9(\mu_{u_L} - \mu_{u_L} - \mu_W) + 3(\mu_\nu - \mu_{e_L}) + 8\mu_W = 0 \quad (412)$$

$$\rightarrow -9\mu_W + 3(\mu_\nu - \mu_{e_L}) + 8\mu_W = 0 \quad (413)$$

$$\rightarrow -\mu_W + 3(\mu_\nu - \mu_{e_L}) = 0 \quad (414)$$

Substituting $\mu_\nu = -3\mu_{u_L}$ and $\mu_{e_L} = \mu_\nu + \mu_W$ into the above equation, we get:

$$-\mu_W + 3(-3\mu_{u_L} - \mu_\nu - \mu_W) = 0 \quad (415)$$

$$\rightarrow -\mu_W - 9\mu_{u_L} - 3\mu_\nu - 3\mu_W = 0 \quad (416)$$

$$\rightarrow -4\mu_W - 9\mu_{u_L} - 3\mu_\nu = 0 \quad (417)$$

Substituting $\mu_\nu = -3\mu_{u_L}$ into the above equation, we get:

$$-4\mu_W - 9\mu_{u_L} + 9\mu_{u_L} = 0 \quad (418)$$

$$\rightarrow -4\mu_W = 0 \quad \Rightarrow \quad \mu_W = 0 \quad (419)$$

Substituting $\mu_W = 0$ into the equation $24\mu_{u_L} + 16\mu_\phi - 8\mu_W = 0$, we get:

$$24\mu_{u_L} + 16\mu_\phi = 0 \quad (420)$$

$$\rightarrow \mu_\phi = -\frac{24}{16}\mu_{u_L} = -\frac{3}{2}\mu_{u_L} \quad (421)$$

Now we can express B and L in terms of μ_{u_L} :

$$B = \frac{T^2}{6} [3(\mu_{u_L} + \mu_{d_L}) + 3(\mu_{u_R} + \mu_{d_R})] \quad (422)$$

$$= \frac{T^2}{6} [3(\mu_{u_L} + \mu_{u_L} + \mu_W) + 3(\mu_{u_L} + \mu_\phi + \mu_{d_L} - \mu_\phi)] \quad (423)$$

$$= \frac{T^2}{6} [3(2\mu_{u_L}) + 3(2\mu_{u_L})] = \frac{T^2}{6} \cdot 12\mu_{u_L}. \quad (424)$$

$$L = \frac{T^2}{6} [3\mu_\nu + 3\mu_{e_L} + 3\mu_{e_R}] \quad (425)$$

$$= \frac{T^2}{6} [3(-3\mu_{u_L}) + 3(-3\mu_{u_L} + \mu_W) + 3(\mu_{e_L} - \mu_\phi)] \quad (426)$$

$$= \frac{T^2}{6} [-9\mu_{u_L} - 9\mu_{u_L} + 3\mu_W + 3\mu_{e_L} - 3\mu_\phi] \quad (427)$$

$$= \frac{T^2}{6} [-18\mu_{u_L} + 3\mu_{e_L} - 3\mu_\phi] \quad (428)$$

$$= \frac{T^2}{6} \left[-18\mu_{u_L} + 3(-3\mu_{u_L} + \mu_W) - 3\left(-\frac{3}{2}\mu_{u_L}\right) \right] \quad (429)$$

$$= \frac{T^2}{6} \left[-18\mu_{u_L} - 9\mu_{u_L} + \frac{9}{2}\mu_{u_L} \right] = \frac{T^2}{6} \cdot \left(-\frac{45}{2}\mu_{u_L} \right). \quad (430)$$

Finally, we can calculate the ratio $B/(B - L)$:

$$\frac{B}{B - L} = \frac{\frac{T^2}{6} \cdot 12\mu_{u_L}}{\frac{T^2}{6} \cdot 12\mu_{u_L} - \frac{T^2}{6} \cdot \left(-\frac{45}{2}\mu_{u_L}\right)} = \frac{12\mu_{u_L}}{12\mu_{u_L} + \frac{45}{2}\mu_{u_L}} = \frac{12}{12 + \frac{45}{2}} = \frac{8}{23} \quad (431)$$

Thus, the ratio $B/(B - L)$ is $\frac{8}{23}$.

Problem Set 8 due on May 8 at 11:59pm

Question 1

- (a) If the quantity $\hat{k} \equiv \frac{k}{a^2 T^2} = -10^{-40}$, determine the temperature at which the Universe becomes curvature dominated.

We can still define “today” to be the time when the microwave background temperature has reached the value 2.7 K.

- (b) In this case, however, what is the value of the Hubble parameter (in km/s/Mpc), and
(c) what is the corresponding age of the Universe (in years) today?
(d) How does the scale factor grow with time?
(e) How does the distance to the horizon scale with time?

Answer

- (a)

First, start with the Friedmann equation, which is given by

$$H^2 = \frac{8\pi G\rho}{3} - \frac{k}{a^2} \quad (432)$$

$$= \frac{8\pi G\rho}{3} + 10^{-40} T^2. \quad (433)$$

To be more specific, we are determining when $\frac{8\pi G\rho}{3} = 10^{-40} T^2$. We assume radiation dominating the process, and then this condition is given by

$$\frac{8\pi G\rho_r}{3} = \frac{8\pi}{3M_p^2} \underbrace{\frac{\pi^2 N}{30}}_{\rho_r} T^4 = 10^{-40} T^2 \quad (434)$$

$$\rightarrow T^2 = 10^{-40} \times \frac{90M_p^2}{8N\pi^3} \quad (435)$$

$$\rightarrow T \approx 10^{-20} \times (0.06) \times M_p \approx 0.0072 \text{ GeV} = 7.2 \text{ MeV}, \quad (436)$$

where we assume $N \approx 100$. However, this temperature is below than the QCD confinement scale $\Lambda_{\text{QCD}} \approx 300 \text{ MeV}$. Hence, the degree of freedom N should be modified to 10.75 (only photons, electrons, neutrinos), and the T is re-evaluated as

$$T \approx 22 \text{ MeV}. \quad (437)$$

(b)

First, $T_0 = 2.7 \text{ K} = 2.327 \times 10^{-4} \text{ eV}$, and the current Hubble parameter H_0 is dominated by curvature, which is given by

$$H_0 = \sqrt{\frac{-k}{a^2}} = \sqrt{10^{-40} T^2} = 10^{-20} \times 2.327 \times 10^{-4} \text{ eV} \quad (438)$$

$$\approx 2.327 \times 10^{-24} \text{ eV}. \quad (439)$$

To convert this to km/s/Mpc, we use the conversion factor $1 \text{ eV} \approx 4.69 \times 10^{34} \text{ km/s/Mpc}$:

$$H_0 \approx 2.33 \times 10^{-24} \times (4.69 \times 10^{34}) \quad (440)$$

$$\approx 1.09 \times 10^{11} \text{ km/s/Mpc}. \quad (441)$$

(c)

Since the universe expansion is dominated by curvature source, the Hubble parameter H is proportional to t^{-1} . Hence, the age of the universe is given by

$$t_0 \approx \frac{1}{H_0} = \frac{1}{2.327 \times 10^{-24} \text{ eV}}. \quad (442)$$

Again, to convert this to years, we use the conversion $1 \text{ s} \approx 1.52 \times 10^{15} \text{ eV}^{-1}$:

$$t_0 \approx \frac{4.29 \times 10^{23}}{1.52 \times 10^{15}} \text{ s} \approx 2.82 \times 10^8 \text{ s}. \quad (443)$$

Given that $1 \text{ year} \approx 3.15 \times 10^7 \text{ s}$, the age of the Universe today is:

$$t_0 \approx \frac{2.82 \times 10^8}{3.15 \times 10^7} \text{ years} \approx 9 \text{ years}. \quad (444)$$

(d)

The definition of hubble parameter is given by

$$H = \frac{\dot{a}}{a} = \sqrt{-\frac{k}{a^2}} = \frac{\sqrt{|k|}}{a}. \quad (445)$$

Hence, we can obtain

$$\dot{a} = \sqrt{|k|} \quad (446)$$

$$\rightarrow a(t) = \sqrt{|k|}t + C, \quad (447)$$

where the constant C can be determined by the initial condition $a(t=0) = 0$. Finally, we get

$$a \propto t. \quad (448)$$

(e)

The horizon distance $d_H(t)$ is defined as:

$$d_H(t) = a(t) \int_0^t \frac{dt'}{a(t')}. \quad (449)$$

We split the integral into two parts: radiation dominated and curvature dominated:

$$d_H(t_0) = a_{\text{cur}}(t_0) \left[\int_0^{t_i} \frac{dt'}{a_{\text{rad}}(t')} + \int_{t_i}^{t_0} \frac{dt'}{a_{\text{cur}}(t')} \right], \quad (450)$$

where $a_{\text{rad}} \propto t^{1/2}$, $a_{\text{cur}} \propto t$, and t_i is when the curvature start to dominate. We can assume $a_{\text{rad}}(t) = At^{1/2}$, $a_{\text{cur}} = Bt$, and $At_i^{1/2} = Bt_i \implies A = B\sqrt{t_i}$. Then, we can complete this integral:

$$d_H(t_0) = Bt_0 \left[\frac{2}{A} \sqrt{t_i} + \frac{1}{B} \ln \left(\frac{t_0}{t_i} \right) \right] \quad (451)$$

$$= t_0 \left(2 + \ln \left(\frac{t_0}{t_i} \right) \right) \quad (452)$$

□

Question 2

Consider the equations of motion for the inflation and radiation

$$\dot{\rho}_\phi + 3H\rho_\phi = -\Gamma_\phi\rho_\phi, \quad (453)$$

$$\dot{\rho}_R + 4H\rho_R = \Gamma_\phi\rho_\phi, \quad (454)$$

- (a) Rewrite these equations in terms of $\frac{d}{dt}$ using $\frac{d}{dt} = RH\frac{d}{dR}$.
- (b) When $\Gamma_\phi \ll H$, the right hand side of the 1st equation can be neglected, leading to the simple solution that $\rho_\phi \approx \rho_{\text{end}}(R_{\text{end}}/R)^3$, where ρ_{end} is the scale factor when inflation ends and $\rho_{\text{end}} = \rho_\phi(R_{\text{end}})$. Provide an analytic solution for ρ_R by integrating the differential equation from R_{end} to some $R > R_{\text{end}}$.
- (c) Using the condition $\rho_R(R_{\text{RH}}) = \rho_\phi(R_{\text{RH}})$, to define R_{RH} . Solve for $R_{\text{end}}/R_{\text{RH}}$ (your result will depend on ρ_{end}). Then find an expression for the reheat temperature (in terms of Γ_ϕ) recalling that $\rho_R = \frac{\pi^2}{30}NT^4$.

Answer

Note: I prefer using a as the scale parameter instead of R .

(a)

By using $\frac{d}{dt}$ using $\frac{d}{dt} = aH\frac{d}{da}$, we can obtain

$$\begin{cases} aH\frac{d\rho_\phi}{da} + 3H\rho_\phi &= -\Gamma_\phi\rho_\phi \\ aH\frac{d\rho_R}{da} + 4H\rho_R &= \Gamma_\phi\rho_\phi \end{cases} \quad (455)$$

$$\rightarrow \begin{cases} a\frac{d\rho_\phi}{da} + 3\rho_\phi &= -\frac{\Gamma_\phi}{H}\rho_\phi \\ a\frac{d\rho_R}{da} + 4\rho_R &= \frac{\Gamma_\phi}{H}\rho_\phi \end{cases} \quad (456)$$

(b)

If the $H \gg \Gamma_\phi$, the equation can be simplified as

$$\begin{cases} a \frac{d\rho_\phi}{da} + 3\rho_\phi &= 0 \\ a \frac{d\rho_R}{da} + 4\rho_R &= \frac{\Gamma_\phi}{H} \rho_\phi \end{cases} \quad (457)$$

$$\rightarrow \frac{d\rho_\phi}{\rho_\phi} = -3 \frac{da}{a} \quad (458)$$

$$\rightarrow \ln \left(\frac{\rho_\phi}{\rho_{\text{end}}} \right) = -3 \ln \left(\frac{a}{a_{\text{end}}} \right) \quad (459)$$

$$\rightarrow \rho_\phi = \rho_{\text{end}} \left(\frac{a}{a_{\text{end}}} \right)^{-3} = \rho_{\text{end}} \left(\frac{a_{\text{end}}}{a} \right)^3. \quad (460)$$

For the radiation part, we have the equation

$$a \frac{d\rho_R}{da} + 4\rho_R = \frac{\Gamma_\phi}{H} \rho_\phi, \quad (461)$$

and we know the hubble H parameter is dominated by the inflation, which is given by

$$H^2 = \frac{8\pi\rho_\phi}{3M_p^2} = \frac{8\pi}{3M_p^2} \rho_{\text{end}} \left(\frac{a_{\text{end}}}{a} \right)^3 \quad (462)$$

$$H = \sqrt{\frac{8\pi}{3M_p^2} \rho_\phi} = \sqrt{\frac{8\pi}{3M_p^2} \rho_{\text{end}} \left(\frac{a_{\text{end}}}{a} \right)^3} = H_{\text{end}} \left(\frac{a_{\text{end}}}{a} \right)^{3/2}, \quad (463)$$

$$\rightarrow \frac{\Gamma_\phi}{H} \rho_\phi = \frac{\Gamma_\phi}{H_{\text{end}} \left(\frac{a_{\text{end}}}{a} \right)^{3/2}} \rho_{\text{end}} \left(\frac{a_{\text{end}}}{a} \right)^3 = \frac{\Gamma_\phi \rho_{\text{end}}}{H_{\text{end}}} \left(\frac{a_{\text{end}}}{a} \right)^{3/2} \quad (464)$$

where $H_{\text{end}} = \sqrt{\frac{8\pi}{3M_p^2} \rho_{\text{end}}}$. Then, the radiation part can be simplified as

$$a \frac{d\rho_R}{da} + 4\rho_R = \frac{\Gamma_\phi}{H} \rho_\phi = \frac{\Gamma_\phi \rho_{\text{end}}}{H_{\text{end}}} \left(\frac{a_{\text{end}}}{a} \right)^{3/2}. \quad (465)$$

Multiplying by a^3 on both sides, then we have

$$\underbrace{a^4 \frac{d\rho_R}{da} + 4a^3 \rho_R}_{\frac{d}{da}(a^4 \rho_R)} = \frac{\Gamma_\phi \rho_{\text{end}}}{H_{\text{end}}} (a_{\text{end}})^{3/2} (a)^{3/2} \quad (466)$$

$$\rightarrow a^4 \rho_R \Big|_{\text{end}}^a = \frac{\Gamma_\phi \rho_{\text{end}}}{H_{\text{end}}} (a_{\text{end}})^{3/2} \frac{2}{5} a^{5/2} \Big|_{\text{end}}^a \quad (467)$$

By assuming $\rho_R(a_{\text{end}}) = 0$, the equation can be simplified as

$$\rho_R = \frac{2}{5a^4} \left(\frac{\Gamma_\phi \rho_{\text{end}}}{H_{\text{end}}} (a_{\text{end}})^{3/2} \right) (a^{5/2} - (a_{\text{end}})^{5/2}) \quad (468)$$

$$= \frac{2}{5} \frac{\Gamma_\phi \rho_{\text{end}}}{H_{\text{end}}} \left[\left(\frac{a_{\text{end}}}{a} \right)^{3/2} - \left(\frac{a_{\text{end}}}{a} \right)^4 \right]. \quad (469)$$

(c)

Define $\rho_R(a_{\text{RH}}) = \rho_\phi(a_{\text{RH}})$, which is given by

$$\rho_R = \frac{2}{5} \frac{\Gamma_\phi \rho_{\text{end}}}{H_{\text{end}}} \left[\left(\frac{a_{\text{end}}}{a_{\text{RH}}} \right)^{3/2} - \left(\frac{a_{\text{end}}}{a_{\text{RH}}} \right)^4 \right] = \rho_{\text{end}} \left(\frac{a_{\text{end}}}{a_{\text{RH}}} \right)^3. \quad (470)$$

Assuming $a_{\text{RH}} \gg a_{\text{end}}$, this equation can be simplified as

$$\frac{2}{5} \frac{\Gamma_\phi \rho_{\text{end}}}{H_{\text{end}}} \left(\frac{a_{\text{end}}}{a_{\text{RH}}} \right)^{3/2} = \rho_{\text{end}} \left(\frac{a_{\text{end}}}{a_{\text{RH}}} \right)^3 \quad (471)$$

$$\rightarrow \left(\frac{a_{\text{end}}}{a_{\text{RH}}} \right)^{3/2} = \frac{2}{5} \frac{\Gamma_\phi}{H_{\text{end}}} \quad (472)$$

$$\rightarrow \frac{a_{\text{end}}}{a_{\text{RH}}} = \left(\frac{2}{5} \frac{\Gamma_\phi}{H_{\text{end}}} \right)^{2/3} = \left(\frac{2}{5} \frac{\Gamma_\phi}{\sqrt{\frac{8\pi}{3M_p^2} \rho_{\text{end}}}} \right)^{2/3}. \quad (473)$$

Now $\rho_R(a_{\text{RH}})$ is given by

$$\rho_R(a_{\text{RH}}) = \frac{2}{5} \frac{\Gamma_\phi \rho_{\text{end}}}{H_{\text{end}}} \underbrace{\left(\frac{a_{\text{end}}}{a_{\text{RH}}} \right)^{3/2}}_{\frac{2}{5} \frac{\Gamma_\phi}{H_{\text{end}}}} = \frac{4}{25} \frac{\Gamma_\phi^2 \rho_{\text{end}}}{H_{\text{end}}^2} \quad (474)$$

$$= \frac{4}{25} \frac{\Gamma_\phi^2 \rho_{\text{end}}}{\frac{8\pi}{3M_p^2} \rho_{\text{end}}} \quad (475)$$

$$= \frac{3}{50\pi} \Gamma_\phi^2 M_p^2. \quad (476)$$

Using the formula $\rho_R = \frac{\pi^2}{30} N T^4$, we can have

$$\rho_R = \frac{\pi^2}{30} N T_{\text{RH}}^4 = \frac{3}{50\pi} \Gamma_\phi^2 M_p^2 \quad (477)$$

$$\rightarrow T_{\text{RH}} = \left(\frac{9}{5\pi^3} \frac{\Gamma_\phi^2 M_p^2}{N} \right)^{1/4}. \quad (478)$$

□

Question 3

Let ϕ be the scalar field driving inflation (the inflaton). After inflation, assume that ϕ is oscillating in a quadratic potential given by m_ψ ($V = \frac{1}{2}m_\phi^2\phi^2$) with an initial amplitude given by M_p . Further assume that the decay rate of ϕ is given by $\Gamma_\phi = m_\phi^3/M_p^2$. Let R_{end} be the scale factor when ϕ begins its oscillations.

- Determine the ratio of the scale factor at the time of decay, R_{RH} to R_{end} your result from the previous problem with this specific decay rate.
- After decay, the Universe is radiation dominated. Your solution for the energy density in radiation is no longer valid as the source term in the 2nd equation of problem 2 vanishes. Find the radiation density (as a function of R) for $R > R_{\text{RH}}$.
- Assume now that another scalar field, ψ with mass m_ψ begins its oscillations with amplitude ψ_0 when $m_\psi = H$. Determine the ratio of the scale factor at the time ψ 's begin to oscillate, R_ψ to R_{end} . Your result will depend on whether ψ begins oscillation before or after ϕ decays. Derive this ratio for both cases and give the condition of m_ψ needed for your solution to be valid.

Answer

(a)

From the previous problem, with $\frac{a_{\text{end}}}{a_{\text{RH}}} = \left(\frac{2}{5} \frac{\Gamma_\phi}{H_{\text{end}}}\right)^{2/3} = \left(\frac{2}{5} \frac{\Gamma_\phi}{\sqrt{\frac{8\pi}{3M_p^2}\rho_{\text{end}}}}\right)^{2/3}$, the energy density ρ_ϕ (which is the potential V) and the decay width Γ_ϕ are given by

$$\begin{cases} \rho_{\text{end}} = V = \frac{1}{2}m_\phi^2 M_p^2, \\ \Gamma_\phi = \frac{m_\phi^3}{M_p^2} \end{cases} \quad (479)$$

$$\rightarrow \frac{a_{\text{end}}}{a_{\text{RH}}} = \left(\frac{2}{5} \frac{\frac{m_\phi^3}{M_p^2}}{\sqrt{\frac{8\pi}{3M_p^2} \frac{1}{2}m_\phi^2 M_p^2}}\right)^{2/3} \quad (480)$$

$$= \left(\sqrt{\frac{3}{25\pi}} \frac{m_\phi^2}{M_p^2}\right)^{2/3} \quad (481)$$

$$\rightarrow \frac{a_{\text{RH}}}{a_{\text{end}}} = \left(\sqrt{\frac{\pi}{3}} \frac{5M_p^2}{m_\phi^2}\right)^{2/3}. \quad (482)$$

(b)

Now, after ϕ decaying, the contribution $\frac{\Gamma_\phi}{H}\rho_\phi$ in the ρ_R differential equation will vanish, which is given by

$$a \frac{d\rho_R}{da} + 4\rho_R = \frac{\Gamma_\phi}{H}\rho_\phi = 0 \quad (483)$$

$$\rightarrow \ln(\rho_R)|_{\text{RH}}^a = -4 \ln(a)|_{\text{RH}}^a \quad (484)$$

$$\rightarrow \frac{\rho_R}{\rho_R(a_{\text{RH}})} = \left(\frac{a_{\text{RH}}}{a}\right)^4 \quad (485)$$

$$\rightarrow \rho_R = \rho_R(a_{\text{RH}}) \left(\frac{a_{\text{RH}}}{a}\right)^4 = \frac{3}{50\pi} \Gamma_\phi^2 M_p^2 \left(\frac{a_{\text{RH}}}{a}\right)^4 \quad (486)$$

$$= \frac{3}{50\pi} \frac{m_\phi^6}{M_p^2} \left(\frac{a_{\text{RH}}}{a}\right)^4 \propto a^{-4}. \quad (487)$$

Particularly, after ϕ decaying, the radiation will freely dilute because of the expansion. In other word, the radiation energy density ρ_R is proportional to a^{-4} .

(c)

According to previous problem, the hubble parameter at reheating is given by

$$H_{\text{RH}} = H_{\text{end}} \left(\frac{a_{\text{end}}}{a_{\text{RH}}}\right)^{3/2} = H_{\text{end}} \left(\frac{2}{5} \frac{\Gamma_\phi}{H_{\text{end}}}\right) = \frac{2}{5} \Gamma_\phi = \frac{2}{5} \frac{m_\phi^3}{M_p^2}. \quad (488)$$

Assuming the ψ field begins oscillate **before** the ϕ decay, the ρ_ϕ dominates the expansion source, and the hubble parameter H now is given by

$$H \propto a^{-3/2} \quad (489)$$

$$H = H_{\text{end}} \left(\frac{a_{\text{end}}}{a}\right)^{3/2} \quad (490)$$

$$\rightarrow \frac{a_{\text{end}}}{a} = \left(\frac{H}{H_{\text{end}}}\right)^{2/3} \quad (491)$$

$$\rightarrow \frac{a}{a_{\text{end}}} = \left(\frac{H_{\text{end}}}{H}\right)^{2/3}. \quad (492)$$

Hence, when $a = a_\psi$, then $H = m_\psi$, and ψ start to oscillate. The equation is then given by

$$\frac{a_\psi}{a_{\text{end}}} = \left(\frac{H_{\text{end}}}{H}\right)^{2/3} = \left(\frac{\sqrt{\frac{8\pi}{3} \frac{1}{2} m_\phi^2 M_p^2}}{m_\psi}\right)^{2/3} = \left(\sqrt{\frac{4\pi}{3}} \frac{m_\phi}{m_\psi}\right)^{2/3}. \quad (493)$$

This condition is valid when $a_\psi < a_{\text{RH}}$, meaning that $H = m_\psi > H_{\text{RH}} = \frac{2}{5} \Gamma_\phi = \frac{2}{5} \frac{m_\phi^3}{M_p^2}$.

Next, assuming the ψ field begins oscillate **after** the ϕ decay (after reheating), the ρ_R dominates the expansion source, and the hubble parameter H now is given by

$$H \propto a^{-2} \quad (494)$$

$$H = H_{\text{RH}} \left(\frac{a_{\text{RH}}}{a} \right)^2 \quad (495)$$

$$\rightarrow \frac{a_{\text{RH}}}{a} = \left(\frac{H}{H_{\text{RH}}} \right)^{1/2} \quad (496)$$

$$\rightarrow \frac{a}{a_{\text{RH}}} = \left(\frac{H_{\text{RH}}}{H} \right)^{1/2}. \quad (497)$$

Hence, when $a = a_\psi$, then $H = m_\psi$, and ψ start to oscillate. The equation is then given by

$$\frac{a_\psi}{a_{\text{RH}}} = \left(\frac{H_{\text{RH}}}{m_\psi} \right)^{1/2} \quad (498)$$

$$\rightarrow \frac{a_\psi}{a_{\text{end}}} = \frac{a_\psi}{a_{\text{RH}}} \frac{a_{\text{RH}}}{a_{\text{end}}} = \left(\frac{H_{\text{RH}}}{m_\psi} \right)^{1/2} \left(\sqrt{\frac{\pi}{3}} \frac{5M_p^2}{m_\phi^2} \right)^{2/3}. \quad (499)$$

This condition is valid when $a_\psi > a_{\text{RH}}$, meaning that $H = m_\psi < H_{\text{RH}} = \frac{2}{5}\Gamma_\phi = \frac{2}{5}\frac{m_\phi^3}{M_p^2}$. □